

RMIT BUSINESS ANALYTICS CHAMPION SEASON 6

ROUND 2: CASE STUDY

TEAM NAME: Sotahami

TEAM MEMBERS: Cao Hoang Son
Ha Duc Tam
Trinh Nhat Minh
Nguyen Thi Ngan Ha

EXECUTIVE SUMMARY

SITUATION

Colgate-Palmolive's toothbrush production line has faced inconsistent operational efficiency, with fluctuating OEE performance and frequent downtime events.

The two sub-indexes Availability and Performance are very low at only about 82% and 70%, pulling the overall OEE down.

Operational performance index OEE is very poor at only about 60% in the past 6 months

COMPLICATION

Error code 02 Quality - Rework due to input material quality accounts for more than 1/5 of total downtime

Top 4 error codes 11,13,14 and 17 caused by machinery and equipment causing a large amount of downtime

Production line operators may misjudge production rates because they do not rely on bottleneck processes.

CHALLENGE

Analyze the impact of errors on the entire production line qualitatively and propose 3 data-driven strategies to fix them while simultaneously measuring the expected results

ANSWERS

RCCP (Rough-Cut Capacity Planning) to find and continuously improve bottlenecks to gradually increase capacity over time

IQC (Incoming Quality Control) to ensure the highest quality of input materials to minimize the occurrence of errors 02 Quality - Rework

Spare part maintenance strategy to improve each engine part in the machine system to minimize the recurrence of errors 11, 13, 14 and 17

GOAL ACHIEVEMENT

The expected OEE average rise from 60% to around 80%, which shows that the entire production system has improved significantly.

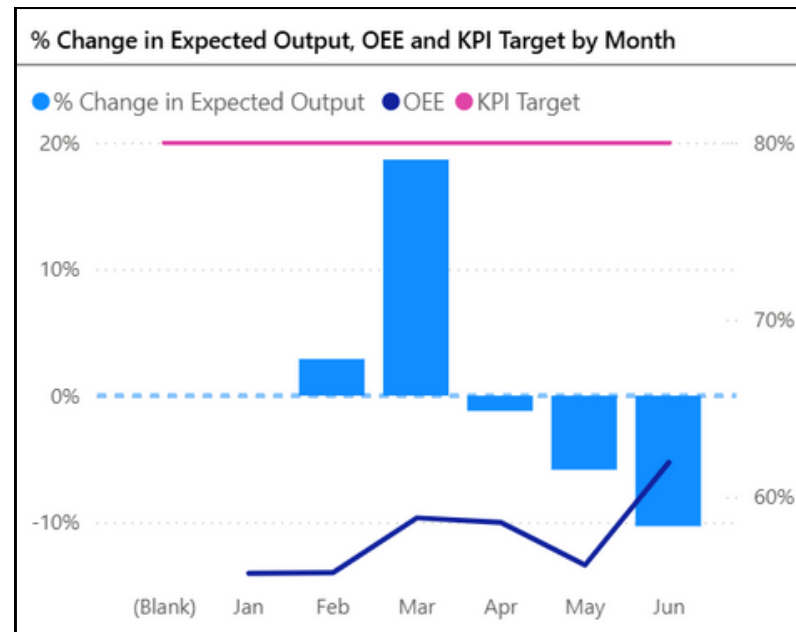
Average Capacity increased by 30%, increasing the Performance index from below 70% to 85%

The ratio of downtime to production availability increased sharply from an average of 82% to 88%.

OVERALL PERFORMANCE

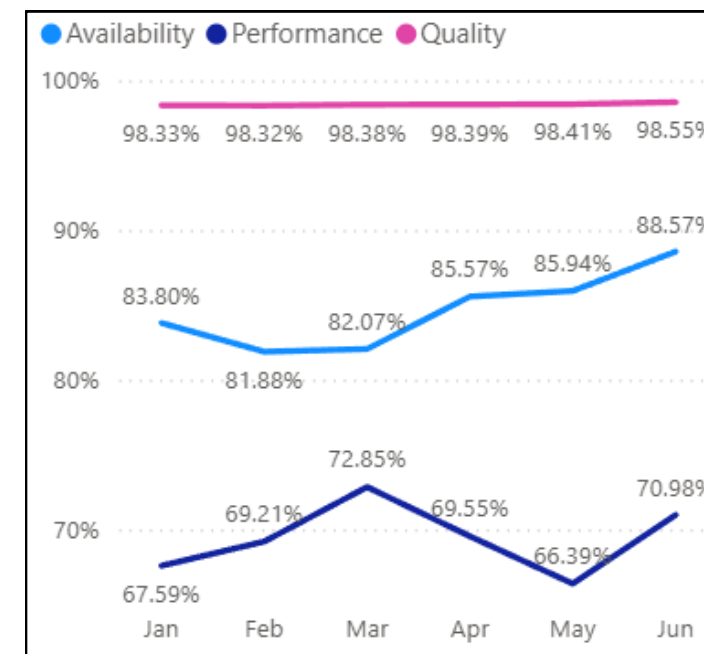
LOW OEE OF 50-65% STEMS FROM WEAK AVAILABILITY AND PERFORMANCE.

Colgate's expected output was consistently behind KPI



From April, OEE improves despite declining **Expected Output**, evidencing genuine efficiency gains. However, OEE remains well below the **80% benchmark**, underscoring an ongoing performance gap driven primarily by **availability losses**

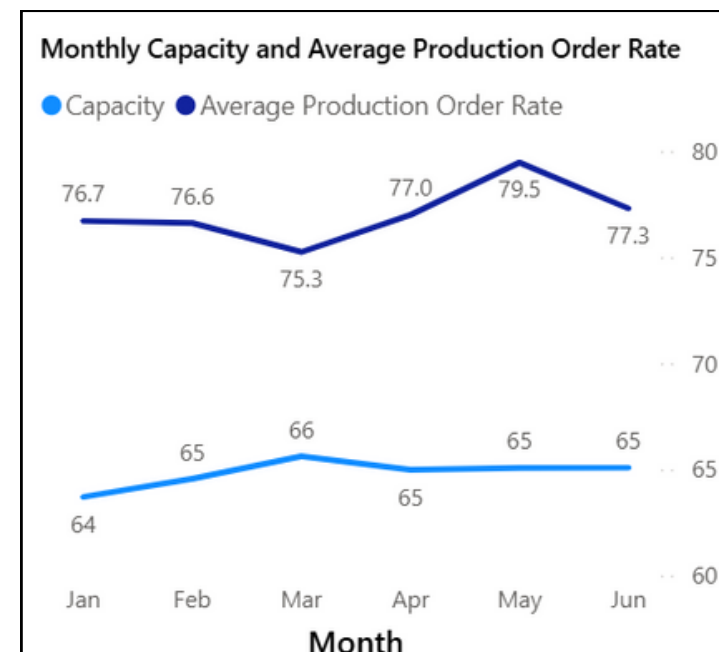
...and this overall gap in OEE stems from low availability & performance



OEE = Availability x Performance x Quality

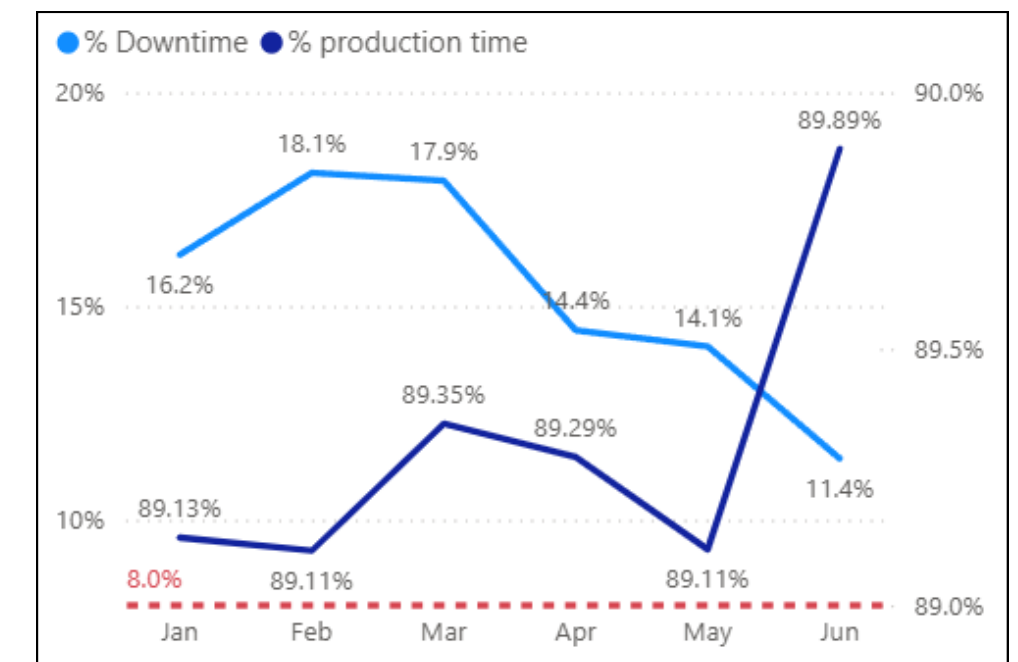
The **Quality** rate has **remained stable** over time, consistently hovering around **98%**. In contrast, the other **two components** contributing to **OEE Availability** and **Performance** are clearly fluctuating and remain at **relatively low levels**. Therefore, in order to **improve overall OEE**, it is essential to focus on **enhancing** both **Availability** and **Performance**.

- The average capacity of the **first two quarters** was only **65 products** per minute compared to the requirement of **75**
- A difference of **10 products** per minute is a huge amount for a production line like Colgate-Palmolive.
- This may explain the **weakness** of **Performance**, the main factor that **drags down OEE**.



Unexpected production downtime occurred more frequently than expected, impacting efficiency

- The **Downtime** and **Production Available Time** ratio is around **15%** while the good number is only around **8%**.
- The **Planned Stops** magnitude index is around **90%**, which is **quite good** considering the **average benchmark** is around **92%**



OEE COMPONENTS ARE LOW DUE TO HIGH UNPLANNED DOWNTIME AND LOW CAPACITY

Colgate-Palmolive's Availability and Performance metrics can be attributed to its capacity and runtime

$$\text{Availability} = \frac{\text{Runtime}}{\text{Planned Production Time}} \quad (\text{Runtime} = \text{Planned Production Time} - \text{Downtime})$$

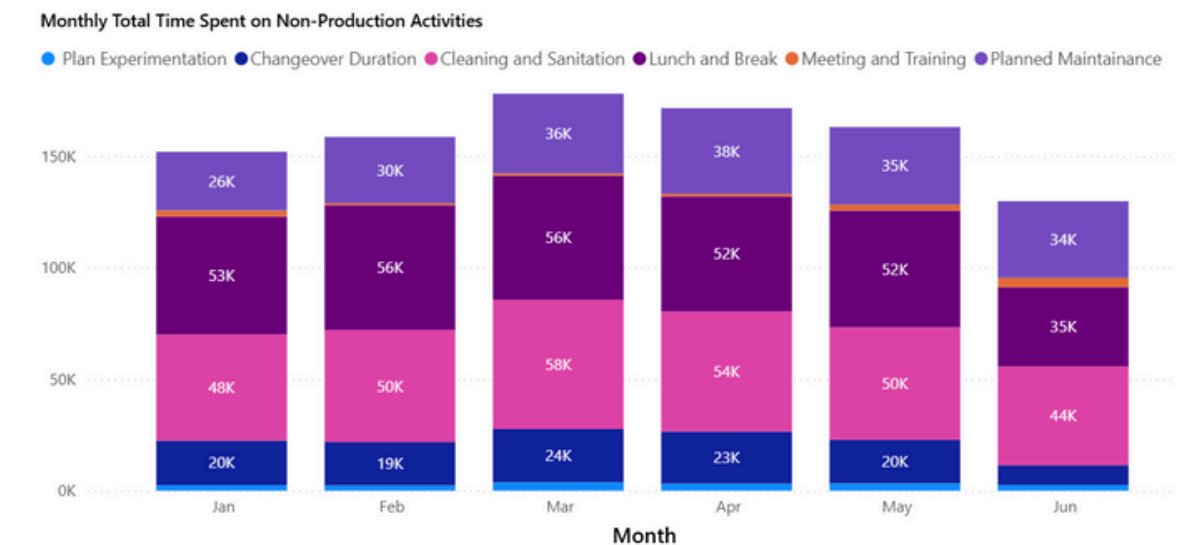
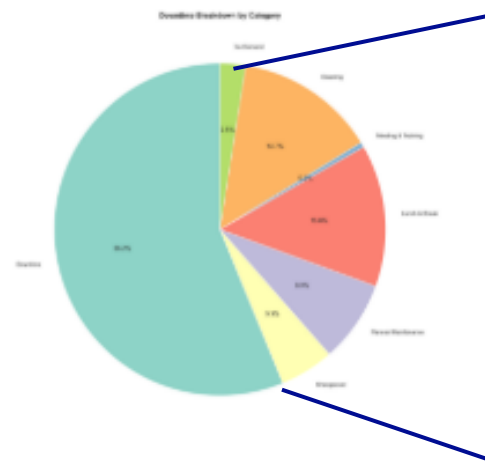
$$\text{Performance} = \frac{\text{Total Production}}{\text{Total Target Production}} \quad (\text{Total Production} = \text{Average Capacity} * \text{Runtime})$$

$$\text{Quality} = \frac{\text{Good Production Quantity}}{\text{Total Production Quantity}}$$

There are 2 Variables need to focused to improve the overall OEE: **Unplanned Downtime & Average Capacity**

- In order to increase **Availability**, the **production line** will need to focus on **reducing downtime** by finding **downtime-related root causes**, so that the ratio between **Runtime** and **Production Available Time** is closer to 100%.
- And to increase the **Performance** index, the factory needs to focus on finding root cause to increase the **average capacity** of the **entire production line**.

Unplanned downtime exceeded the planned downtime



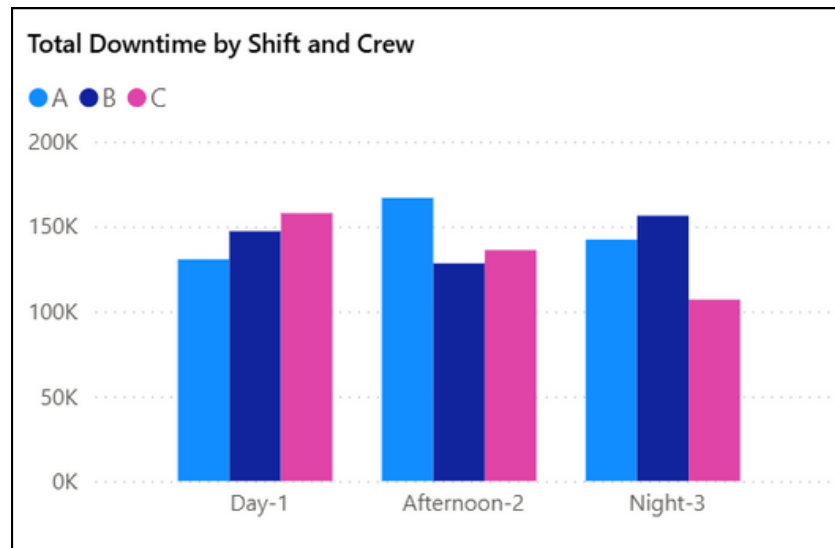
Key Consideration: Unplanned Downtime exceeding Planned Downtime affects manufacturer availability, making a large gap between actual and target production

By **improving capacity and unplanned downtime**, we can raise Availability and Performance, which in turn raise Colgate-Palmolive overall OEE

ROOT CAUSE ANALYSIS - DOWNTIME

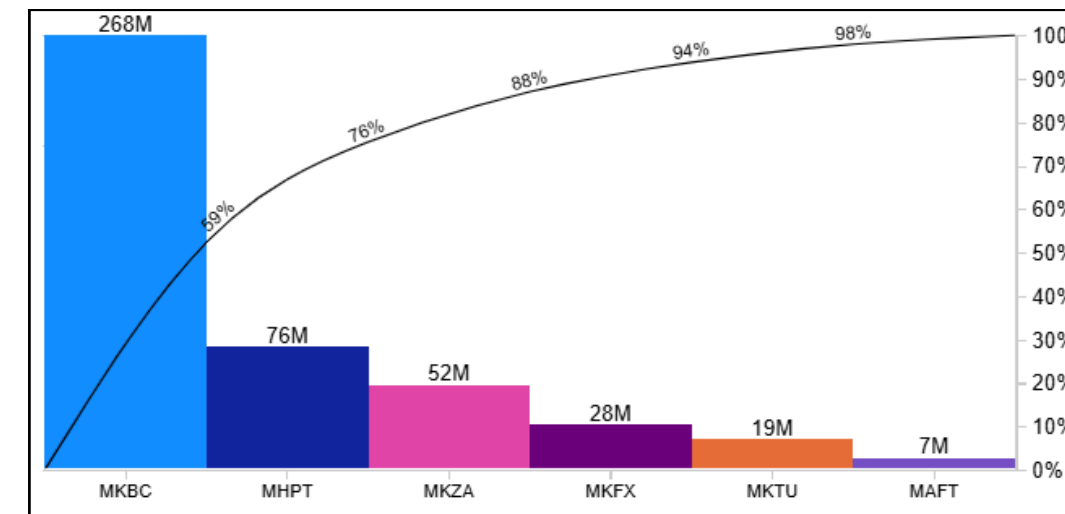
LINES MKBC AND MKZA DRIVE MOST OUTPUT AND 70% OF TOTAL UNPLANNED DOWNTIME

While unplanned downtime does not vary much for crew and shift...



Downtime levels are **relatively consistent across all shifts and crews**, indicating that neither crew assignment nor shift timing significantly impacts overall equipment downtime.

It varies significantly by Production Line Group, especially MKBC and MKZA



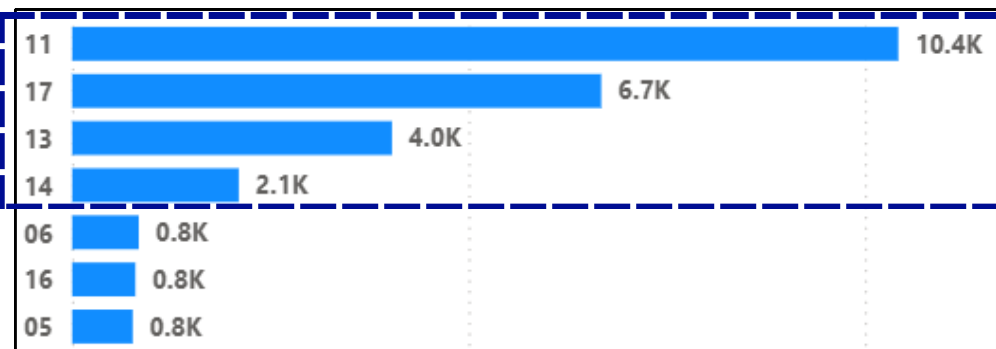
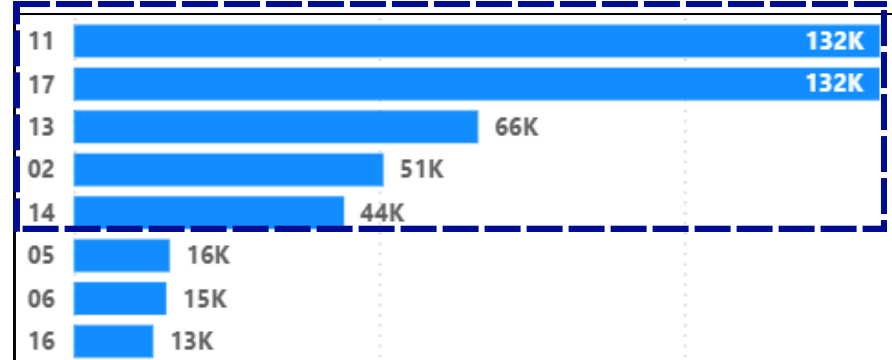
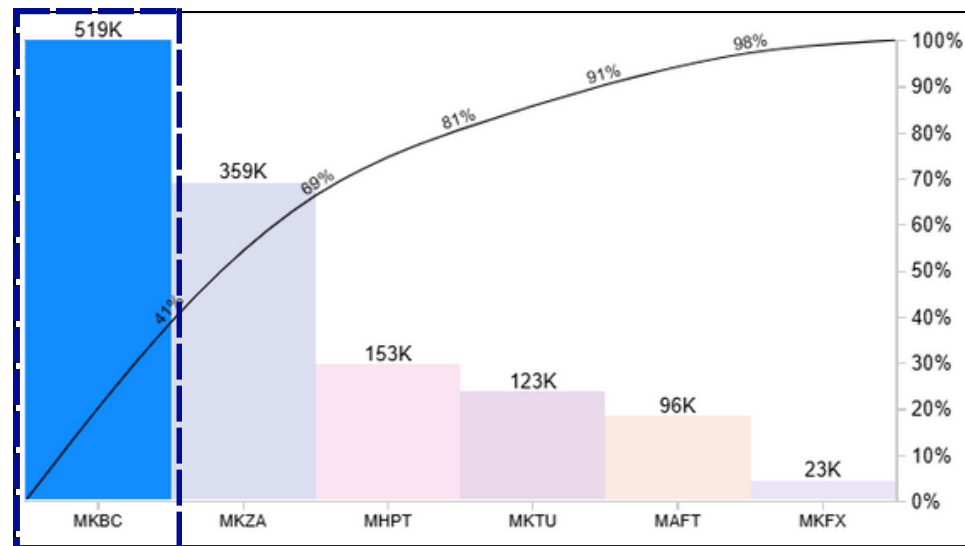
Line groups **MKBC** and **MKZA** contribute the **majority of total output** but also account for a **significant portion of downtime**

The MKBC and MKZA lines are identified as key groups that should be prioritized for a deeper analysis of unplanned downtime causes

ROOT CAUSE ANALYSIS - DOWNTIME

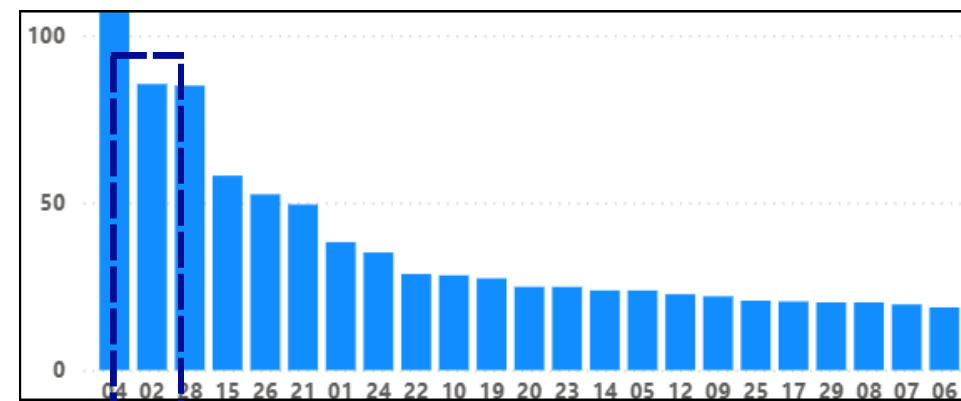
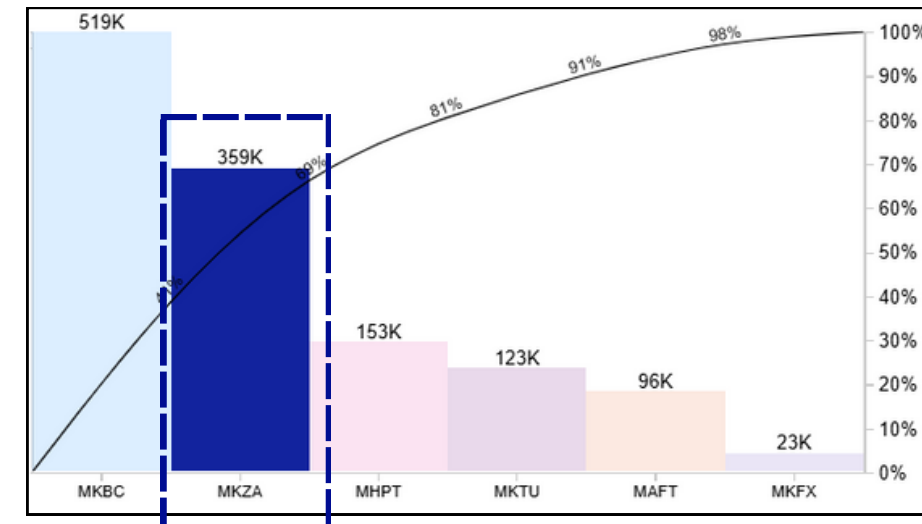
IN ORDER TO FIND THE **ROOT CAUSES** OF LARGE DOWNTIME, WE WILL LOOK AT THE DOWNTIME FOR EACH **ERROR CODE** AND SEE WHICH LINE FAMILIES HAVE THE **HIGHEST DOWNTIME**.

MKBC line family has 5 main errors: 02, 11, 13, 14 and 17. In which the 4 error codes behind have the highest error rate.



- In the chart of total downtime of MKBC, it can be seen that **4 error codes 11,13, 14 and 17** have the highest number of recurrences
- Meanwhile, the average downtime of these 4 codes is not high.
- This proves that the 4 error codes 11,13,14 and 17 are contributing to the large total downtime due to the high number of recurrences on the MKBC line.

The MKZA line group mainly encountered **Quality problems (Error Code 02)**, with high average time per error

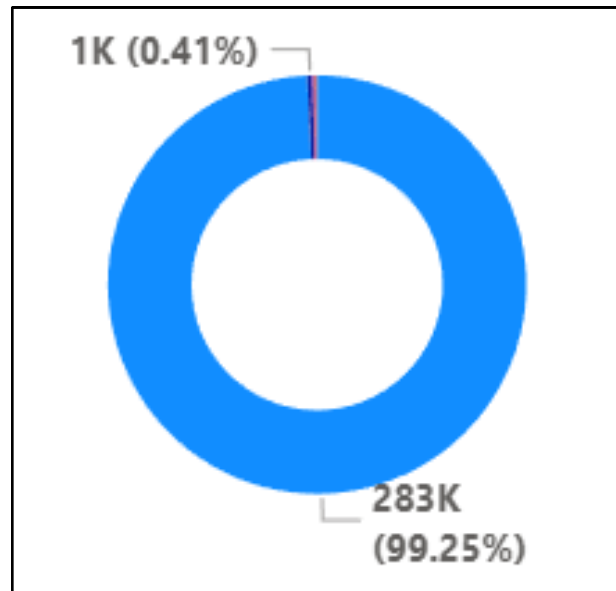


- Meanwhile, the total downtime in MKZA line is currently the highest at error 02 - Quality.
- The downtime of this error code 02 is very high, about 100 minutes each time while the number of downtimes is not equal to the other 4 errors
- This proves that the large amount of downtime in MKZA line is due to the high average downtime of error 02 and the relatively high number of occurrences

ROOT CAUSE ANALYSIS - DOWNTIME

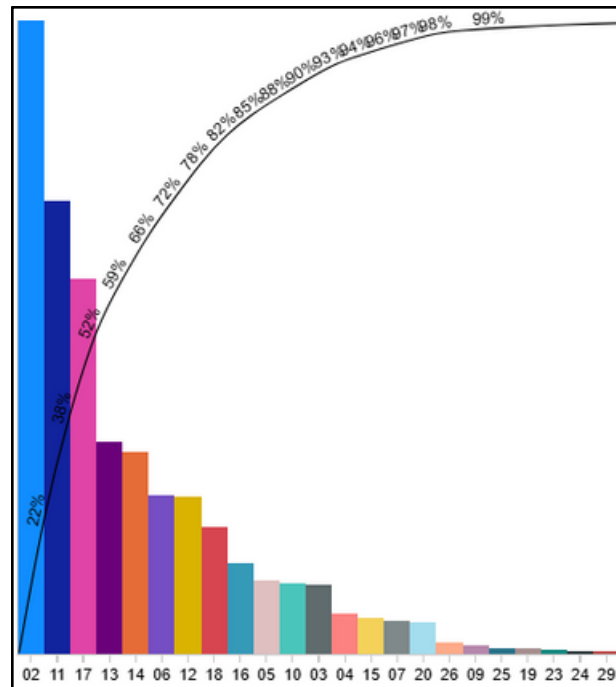
ERROR CODE 02 QUALITY - REWORK IS MAINLY DUE TO THE QUALITY OF INPUT MATERIALS, WHILE 1 1, 1 3, 1 4 AND 1 7 ARE DUE TO THE AMOUNT OF MACHINERY.

Reason code 0202 - Quality - Rework is the most serious factor that contributing to the whole Downtime



Reason code 02 - Quality - Rework contributes to 99% of total downtime of 02 error family code and 22% of total Downtime

- **Time-consuming recovery:** Each rework incident requires stopping the line, cleaning, product segregation, and reprocessing, leading to **long recovery times and lower availability.**
- **Root causes:** Often caused by **filling and sealing defects, misalignment, or unstable process parameters**, indicating weaknesses in process control and equipment precision.



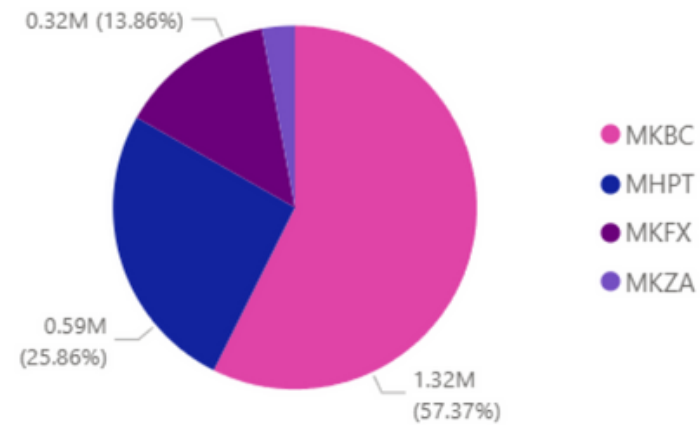
Most downtime is caused by mechanical and alignment failures, with electrical issues and rework loops further reducing line efficiency.

Main Error Group	Typical Examples	Root Causes	Main Impacts
1 Mechanical - Misalignment / Wear & Tear (≈ 60-70%)	Handle stuck on chain, robot position offset, loose screws, worn cutter or welding tip	Natural wear, insufficient maintenance, improper assembly, low mechanical precision	Frequent stoppages, manual adjustment required, lower equipment efficiency
2 Electrical / Control - Sensor or Program Failure (≈ 20%)	X-Y motor failure, encoder error, sensor malfunction, robot program fault	Faulty electrical parts, loose wiring, poor synchronization between mechanical & control systems	Automation errors, unplanned downtime, process instability
3 Cleaning / Re-inspection - Rework Loops (≈ 10%)	Filler cleaning, post-rework inspection, repeated testing	Complex rework process, lack of quick inspection standard	Increased downtime, reduced Availability & Performance (OEE loss)

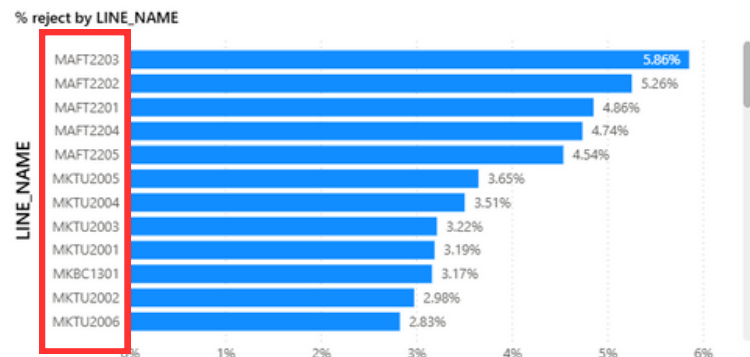
TARGETED QUALITY CONTROL IS REQUIRED FOR BOTH HIGH-VOLUME AND HIGH-REJECT LINES.

Specifically, reducing Reject Products Rate in small volumn lines, such as MAFT and MKTU

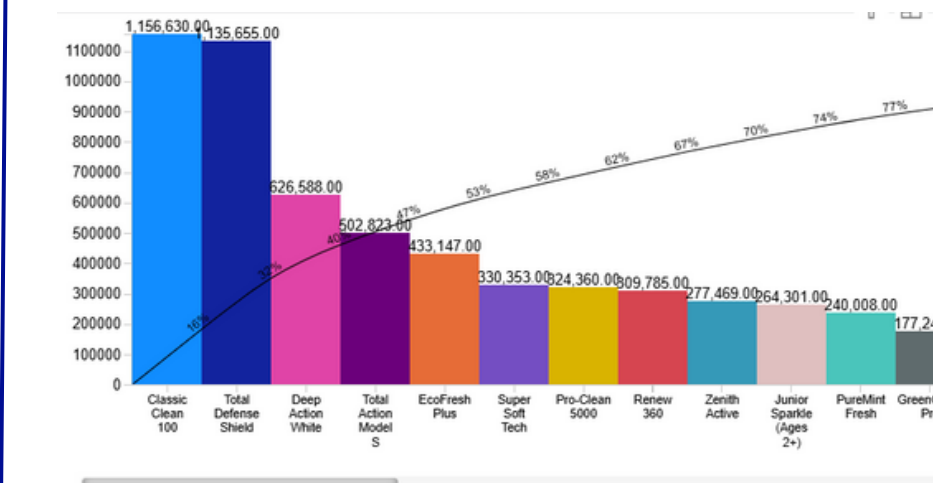
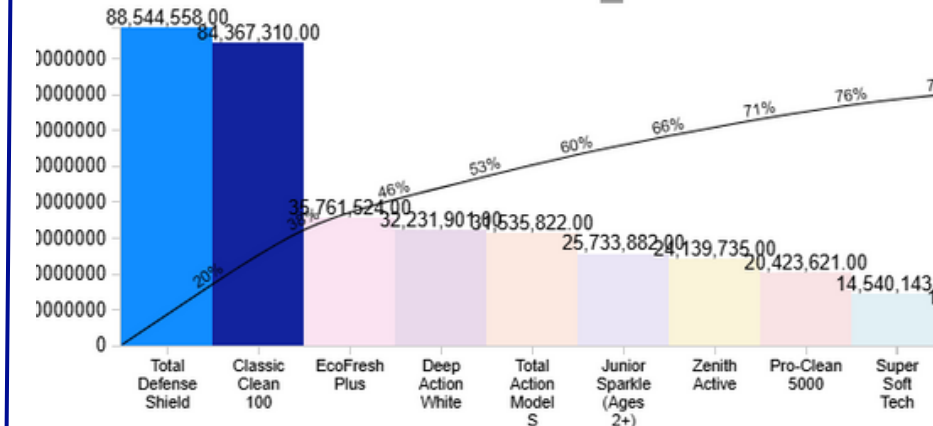
Sum of REJECT_PRODUCTION_QTY by Line code



MKBC & MHPT lines produce the most and have the highest defect rates → Key focus for quality improvemen



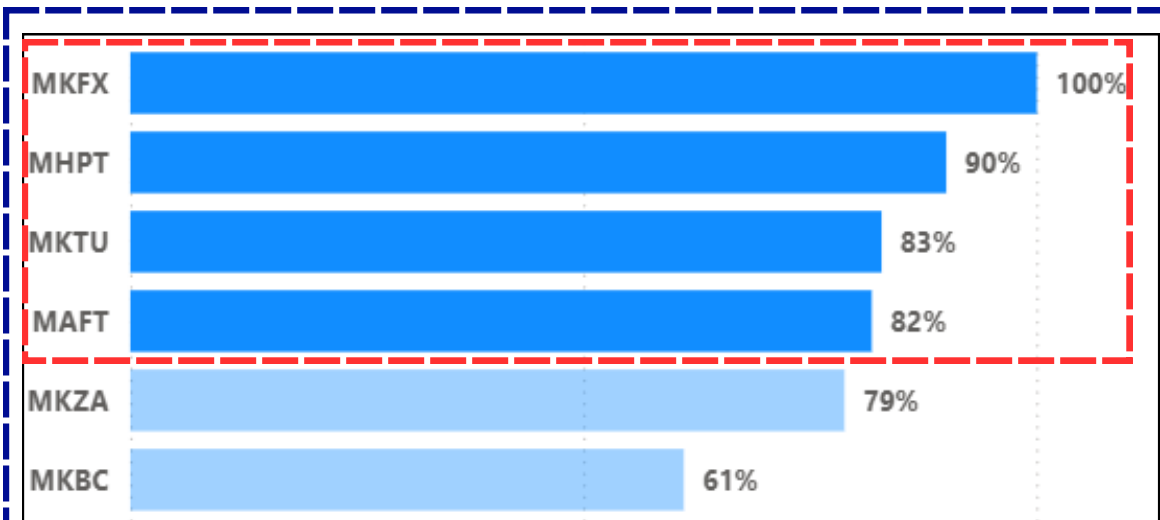
...and enhancing quality control in high-volume lines such as MKBC, MHPT, and MKZA



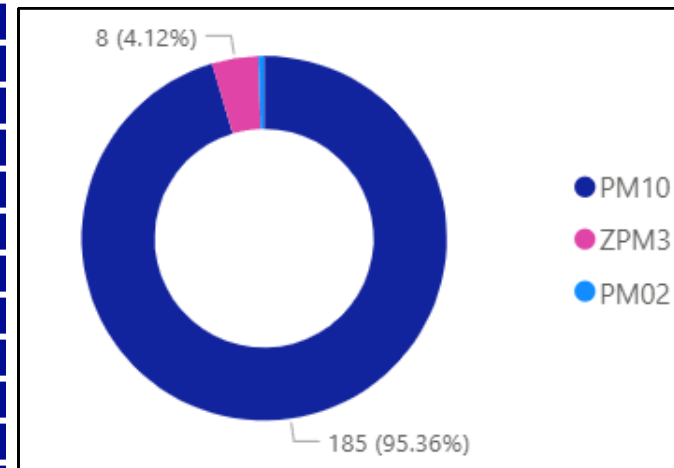
When breaking down defects by product line, the **first two product groups** account for about **40% of total output** and **30% of total defects**. It is **Classic Clean 100** and **Total Defense Shield**

PERFORMANCE IS CONSTRAINED BY A SIGNIFICANT GAP BETWEEN TARGET AND ACTUAL CAPACITY.

The top 4 lines with the highest Production Order completion rate have a high number of Runtimes of over 300 minutes per Runtimes

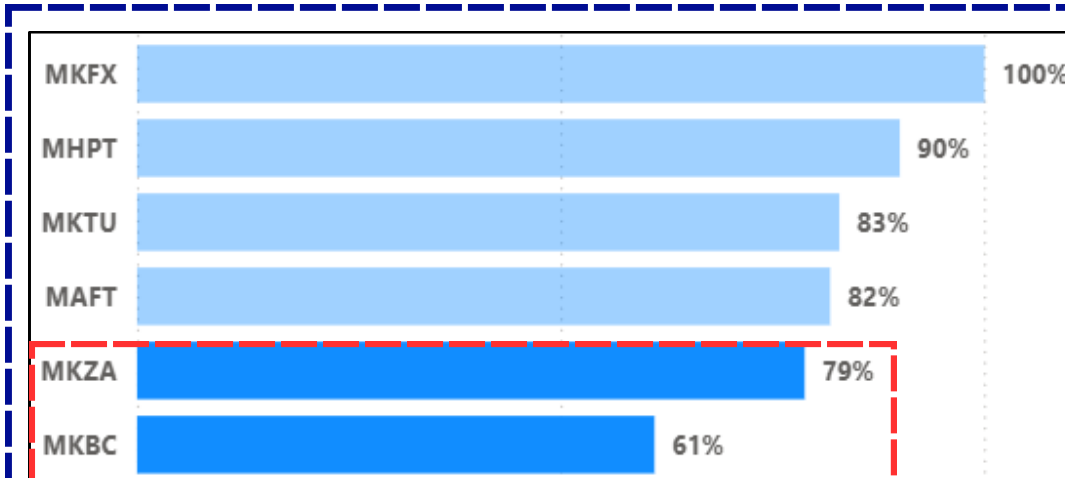


The top 4 lines with the highest completion rate are MKFX, MHPT, MKTU and MKFT, which have a lot of Runtime running over 300 minutes.

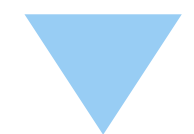


In the pie chart we can see that the entire production system is mainly maintained only when an error occurs and almost never before.

...along with the next two lines with low completion rate which has more number of Runtimes of under 300 minutes per Runtimes.



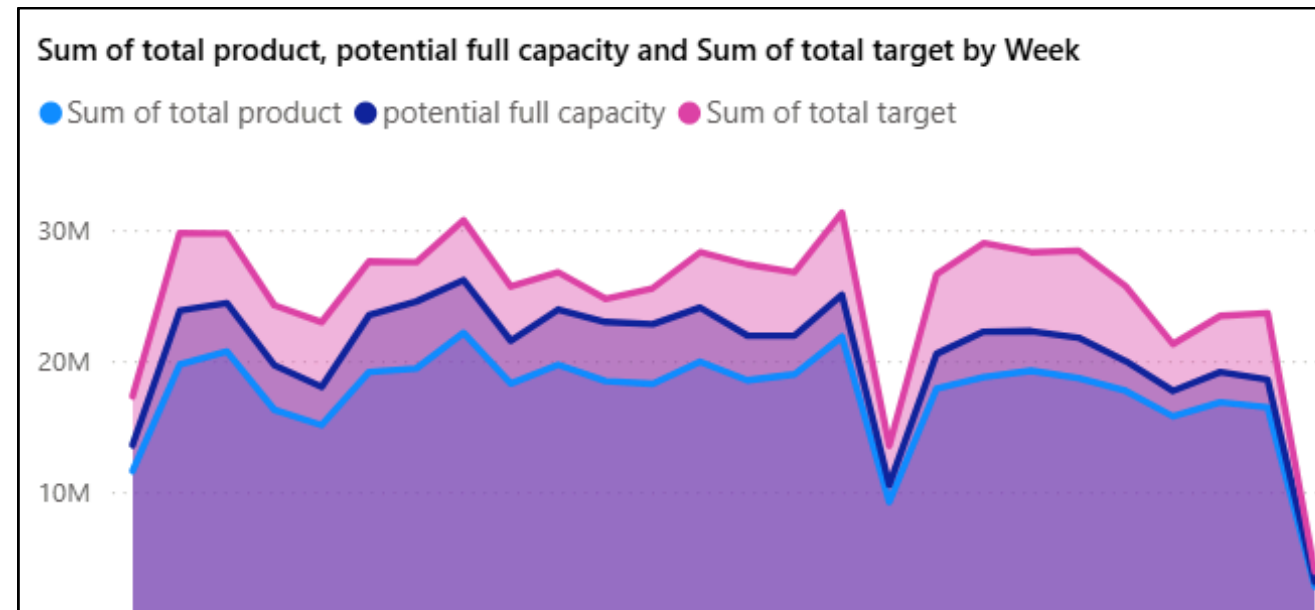
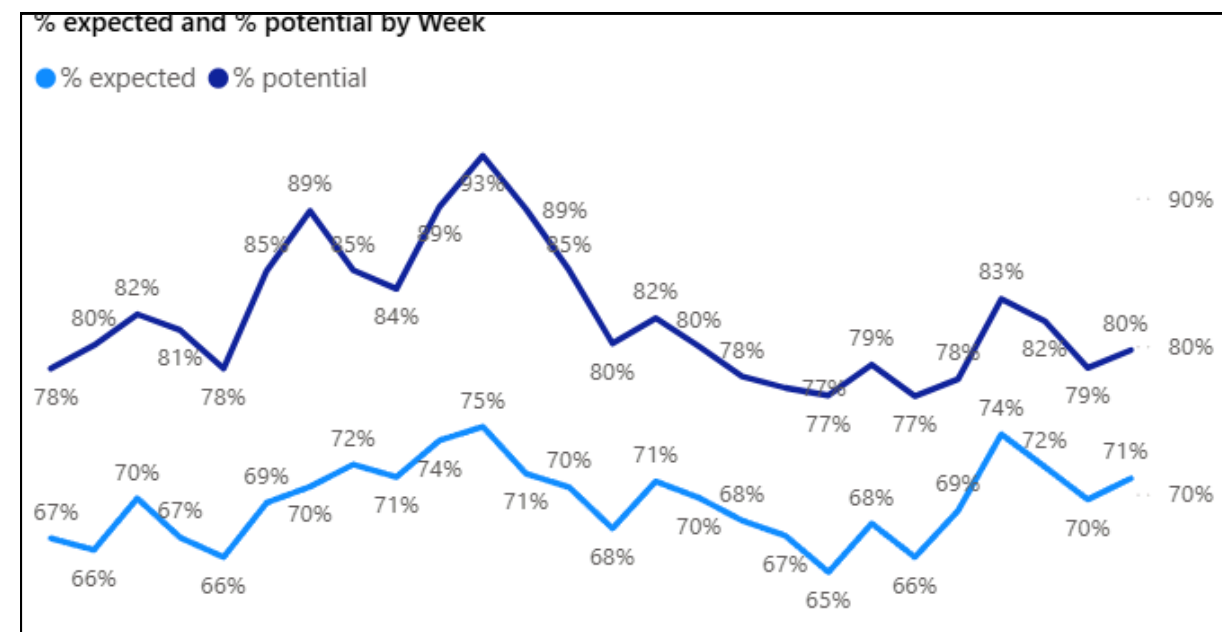
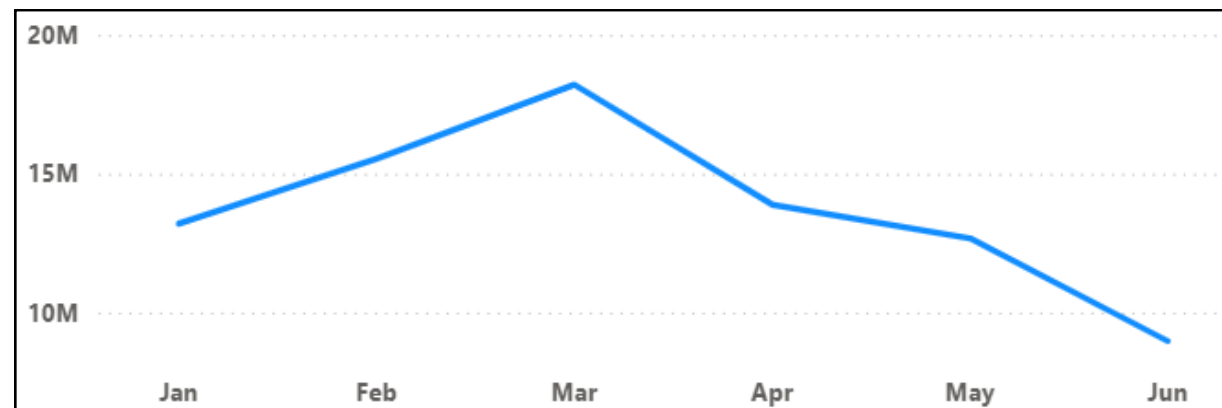
- This proves that the root cause of the failure to achieve the completion rate is due to the lines not achieving enough runtime.
- The reason for the line having to stop many times is due to errors 11, 13, 14 and 17 as analyzed in the previous slide



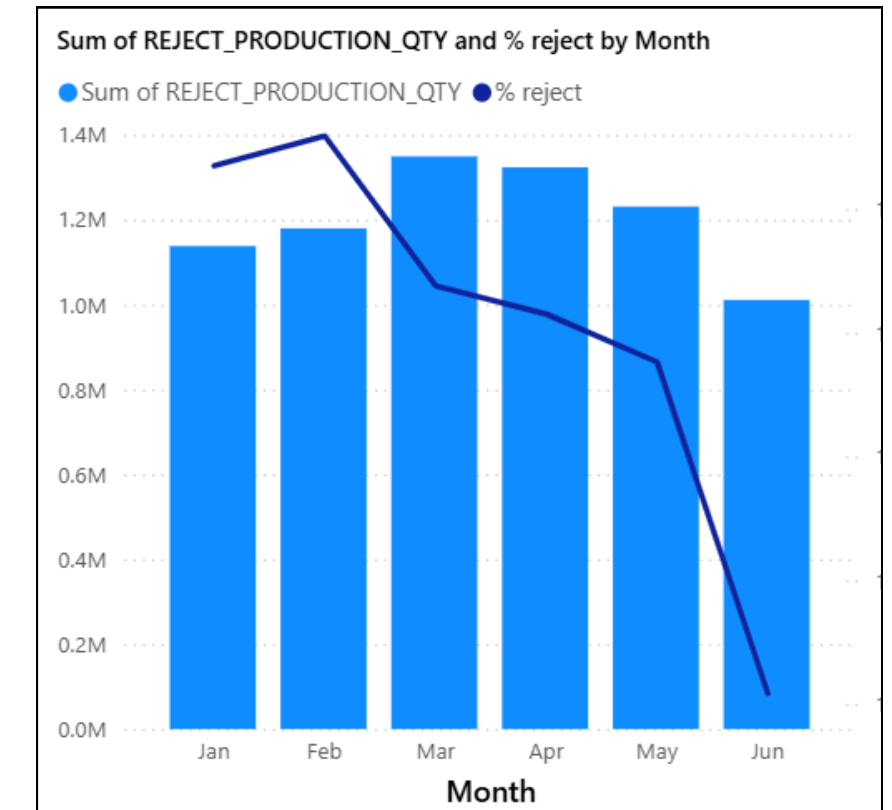
The reason is most likely that the operator may set the target not based on the **bottleneck** and thus overestimate the real capacity.

PERFORMANCE IS CONSTRAINED BY A SIGNIFICANT GAP BETWEEN TARGET AND ACTUAL CAPACITY.

After excluding all downtime durations, we were able to generate a chart illustrating the different between the actual output and the potential output



- The **potential output** can reach nearly 20 million products in March
- On average, **actual output** reached only **85–90%** of potential capacity, indicating a consistent loss of 10–15% due to downtime or inefficiencies. In certain weeks, actual output dropped to **below 60%**, likely due to some problems that cause the downtime
- The gap between actual output and target represents an opportunity for improvement through root cause analysis and better planning or capacity adjustment

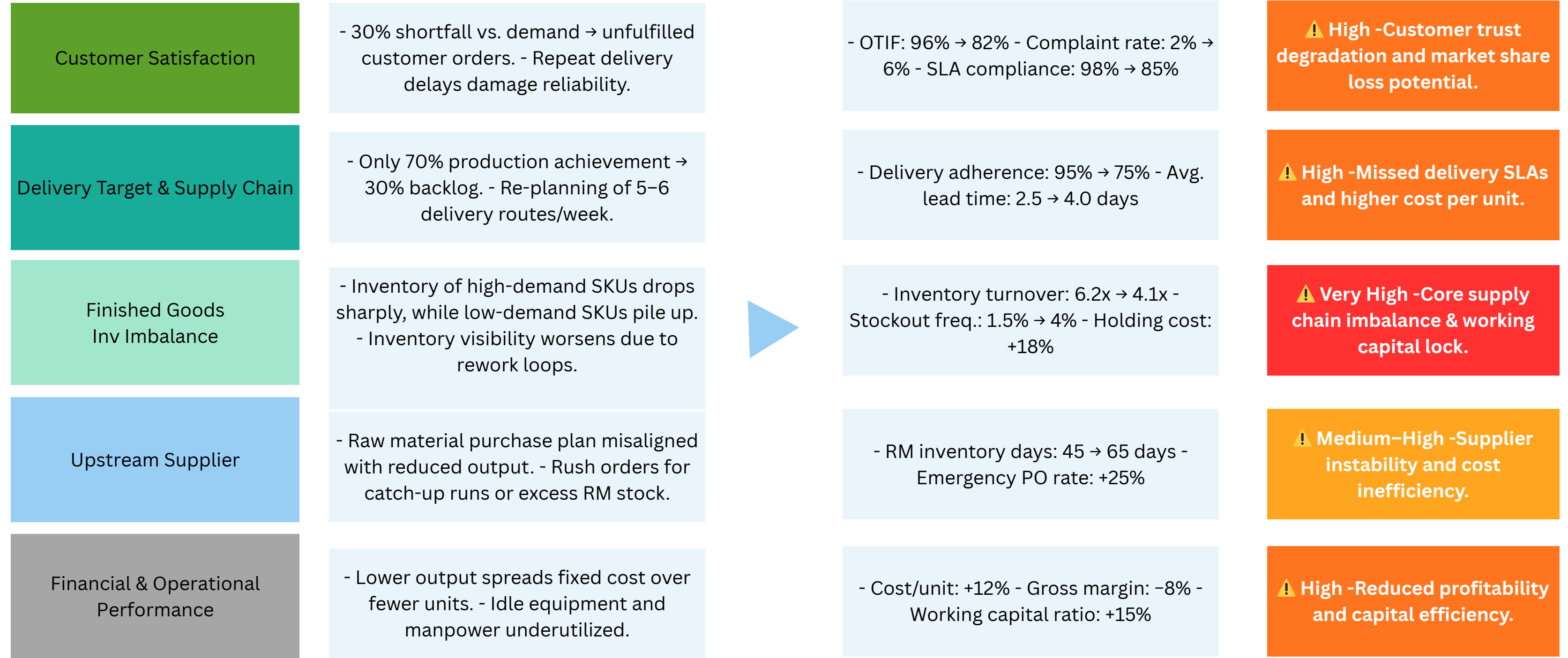


Material Loss Estimation (Defective Units Analysis)

- Total defective units (Jan–Jun): **~ 7.45 million units**
- Estimated material loss: **~ USD 1.12 million** (based on USD 0.15/unit)
- Reject rate decreased from **1.67% → 1.45%**, but total loss remains significant due to high production volume.
- Further **0.1%** reduction could save **~70,000 units/month (~USD 10,500/month)**.
- Focus areas: Process optimization, early defect detection, supplier quality control

BUSINESS IMPACT QUANTIFICATION

OPERATING AT ONLY 70% OF PRODUCTION CAPACITY CAUSES A CHAIN REACTION ACROSS SUPPLY, LOGISTICS, AND FINANCE -CREATING STOCK IMBALANCE IN CUSTOMER DISATISFACTION



STRATEGY 1: INCOMING QUALITY CONTROL

MINIMIZING REWORK-RELATED DOWNTIME (**ERROR 0202**) BY STRENGTHENING MATERIAL QC AT THE INCOMING STAGE, ENSURING ONLY CONFORMING RAW AND PACKAGING MATERIALS ENTER PRODUCTION.

Pillar

Material Qualification & Supplier Control

Enhanced Incoming Sampling & Testing

Incoming-to-Line Verification & Traceability

Cross-Functional Feedback & Continuous Improvement

Objective

Ensure all critical materials meet performance and quality standards before arrival.

Detect non-conforming materials before production.

Prevent defective materials from reaching main lines.

Create a fast, closed feedback loop to eliminate root causes.

Deliverables

- Enforce **Supplier Quality Agreements (SQA)** with stricter specs.
- Require **CoA/CoC** for each lot.
- Conduct **Pre-Shipment Inspection (PSI)** for high-risk suppliers.

- Apply tighter **AQL sampling plans** for key materials.
- Include **functional performance tests** (seal, torque, adhesion).
- Use **dynamic sampling frequency** based on supplier trends.

- Flag high-risk batches in **SAP (Q-status)** until validated.
- Conduct first-off line trials for new/critical lots.
- Reject or regrade materials failing tests.

- Establish **IQC-Production-Supplier** triad reviews for rework.
- Use BD/CAPA reports for corrective actions.
- Share real-time rework dashboards highlighting top defects.

STRATEGY 2: SPARE PART MAINTENANCE

ENSURE MECHANICAL AND ELECTRICAL RELIABILITY THROUGH PROACTIVE SPARE PART MANAGEMENT, MINIMIZING DOWNTIME FROM WEAR, BREAKDOWN, AND UNPLANNED REPAIRS.

Pillar

Spare Part Standardization & Inventory Control

Preventive Maintenance & Replacement Planning

Repair & Refurbishment Optimization

Supplier Collaboration & Continuous Improvement

Objective

Ensure spare parts availability and standard quality across all production lines.

Reduce breakdowns caused by wear and tear or overdue replacement.

Shorten repair time and ensure repaired parts meet functional standards.

Strengthen supplier partnerships to improve part reliability and delivery performance.

Deliverables

- Build critical spare parts list (A/B/C).
- Maintain min stock and apply FIFO to avoid mismatch.

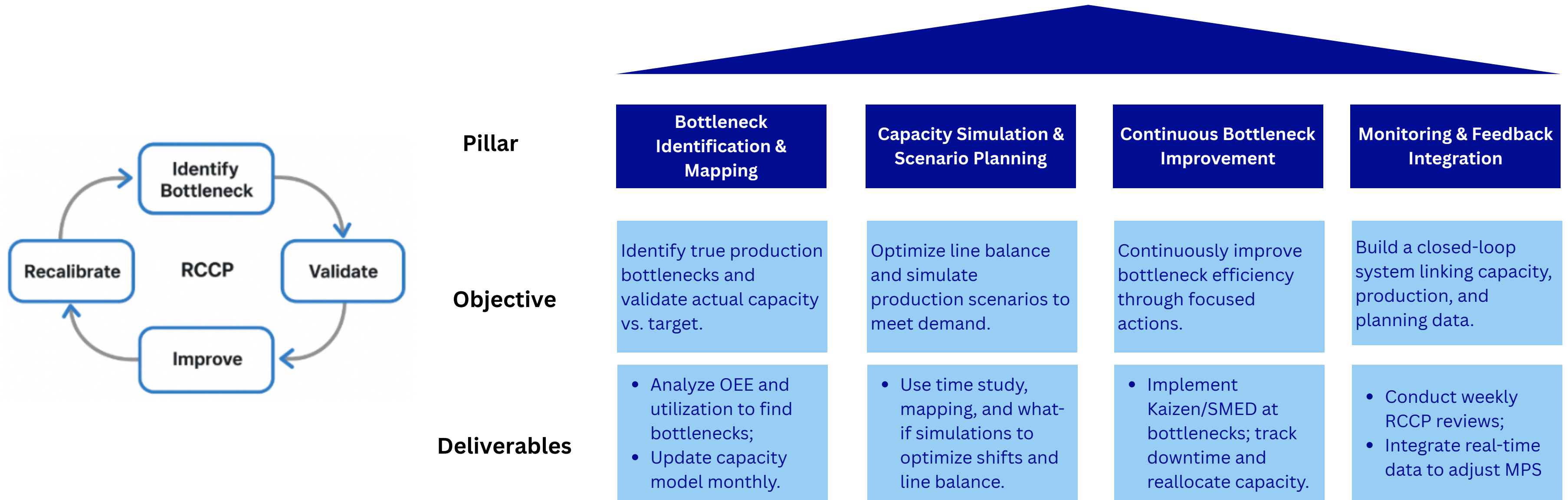
- Set preventive replacement based on runtime/failure data.
- Use condition monitoring to predict wear; plan shutdowns for key machine

- Standardize repair SOPs and history tracking in SAP.
- Compare repair vs. replace for cost and performance.

- Run joint failure analyses with suppliers.
- Require CoA/CoC for refurbished parts; use supplier scorecards on defects and delivery.

STRATEGY 3: RCCP (ROUGH-CUT CAPACITY PLANNING)

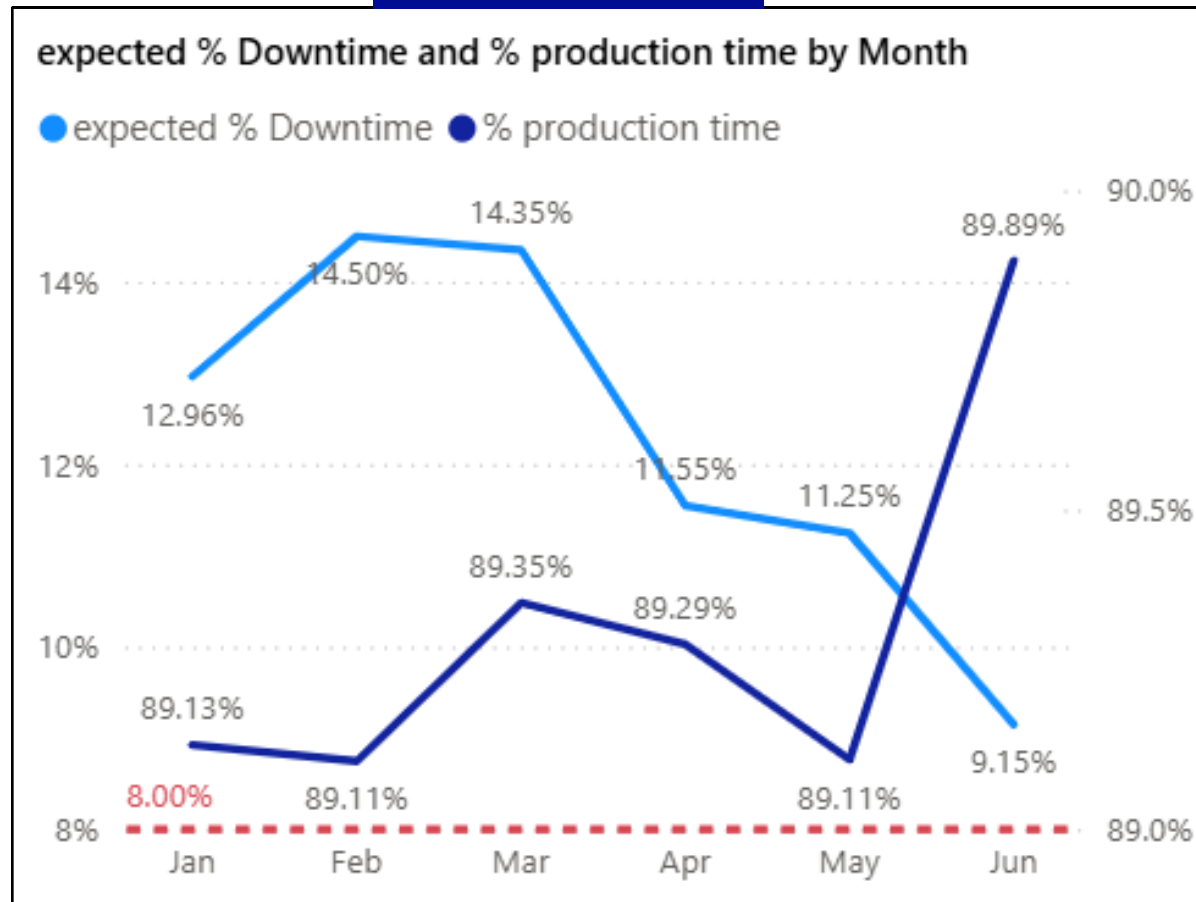
CONTINUOUSLY IDENTIFY AND OPTIMIZE BOTTLENECK RESOURCES TO ALIGN ACTUAL CAPACITY WITH PRODUCTION TARGETS AND REDUCE LOST OUTPUT.



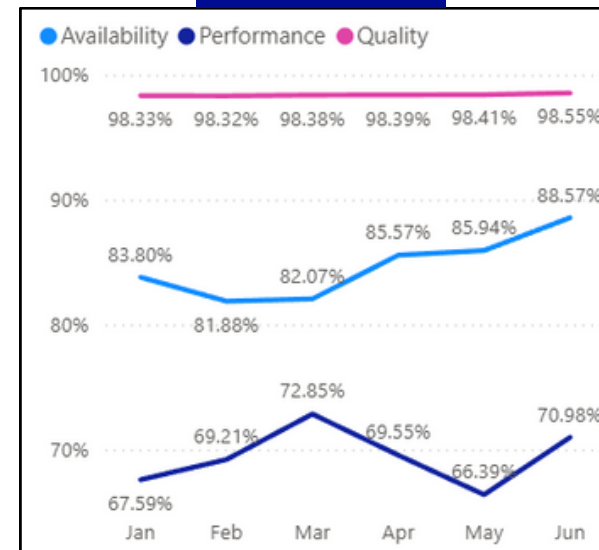
RESULTS AFTER STRATEGY IMPLEMENTATION

A 30% increase in capacity and a 20% reduction in downtime were observed, representing a remarkable improvement in operational efficiency.

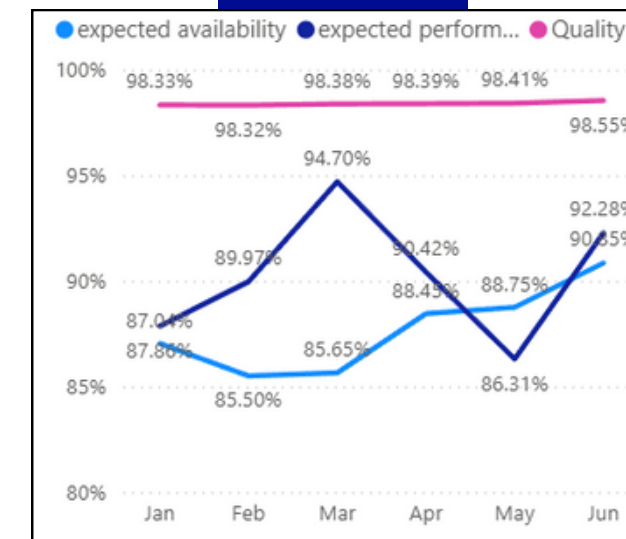
Applying Strategy



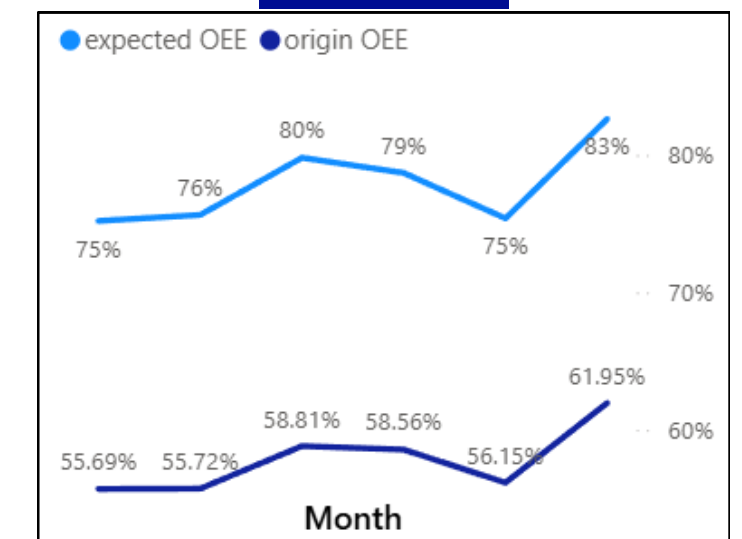
Before



After

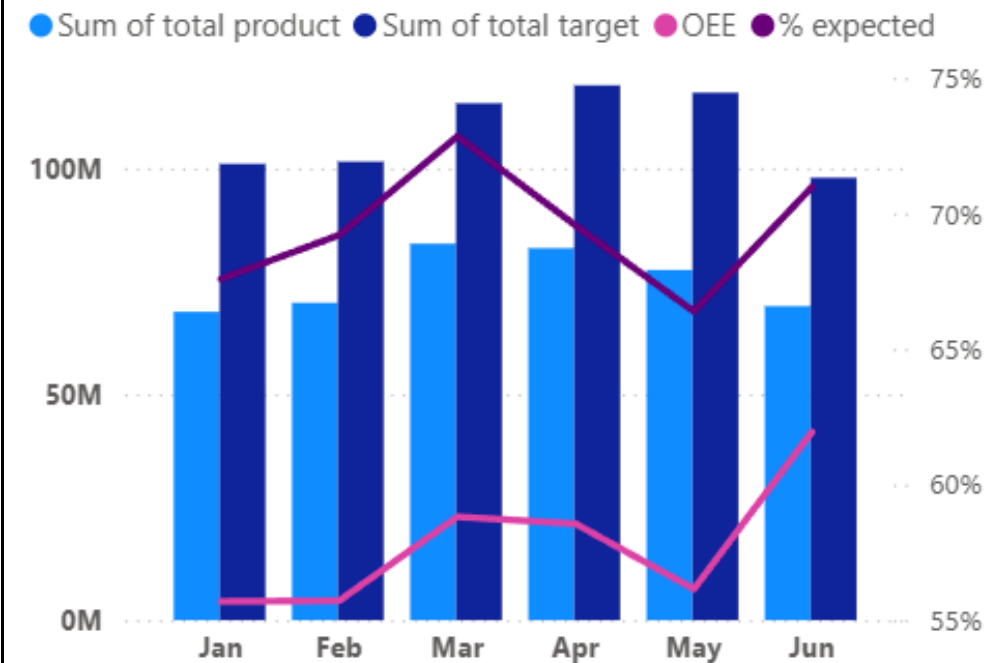


OEE Result

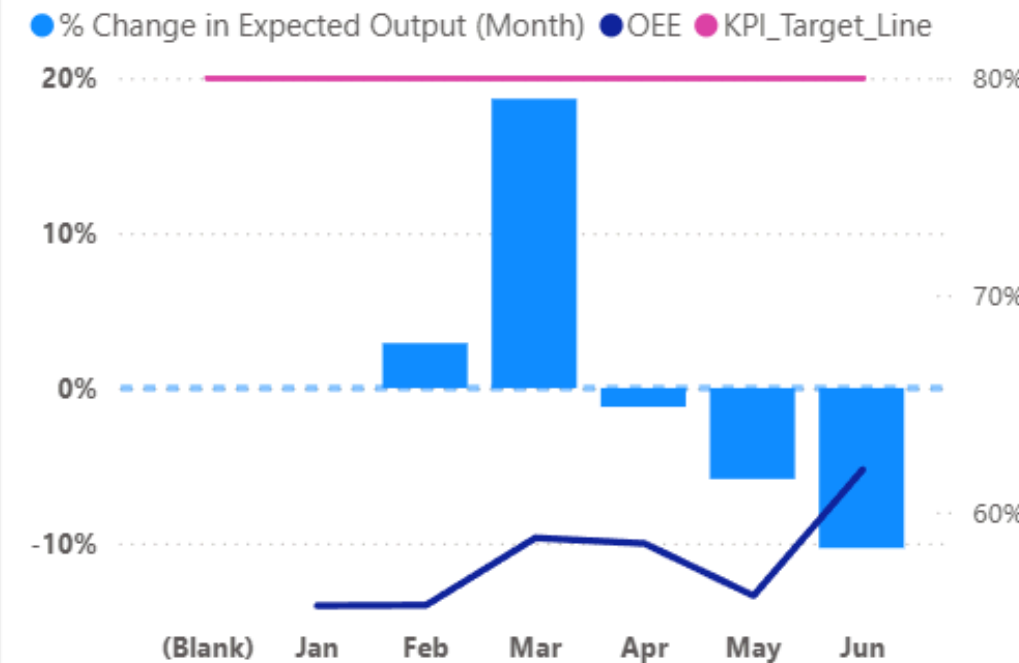


It is clearly observed that, following the implementation of the proposed strategies **Availability** increased significantly by approximately **3–5%** month over month. A notable improvement was also seen in Performance, with a sharp rise of at **least 20%**, which in turn contributed to a **substantial boost in OEE**, enabling it to reach the “**world-class benchmark**” of 80% as initially targeted.

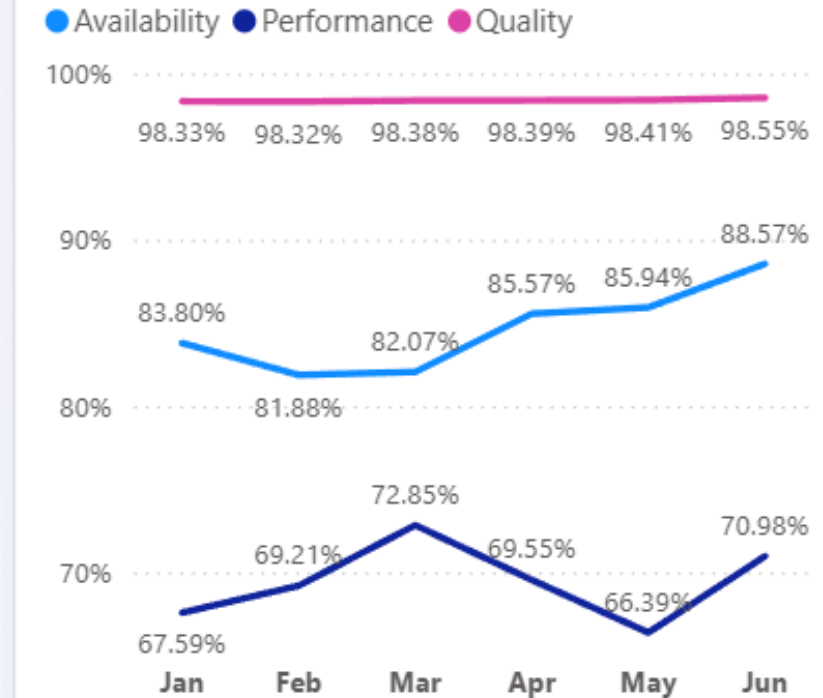
Sum of total product, Sum of total target, OEE and % expected by Month



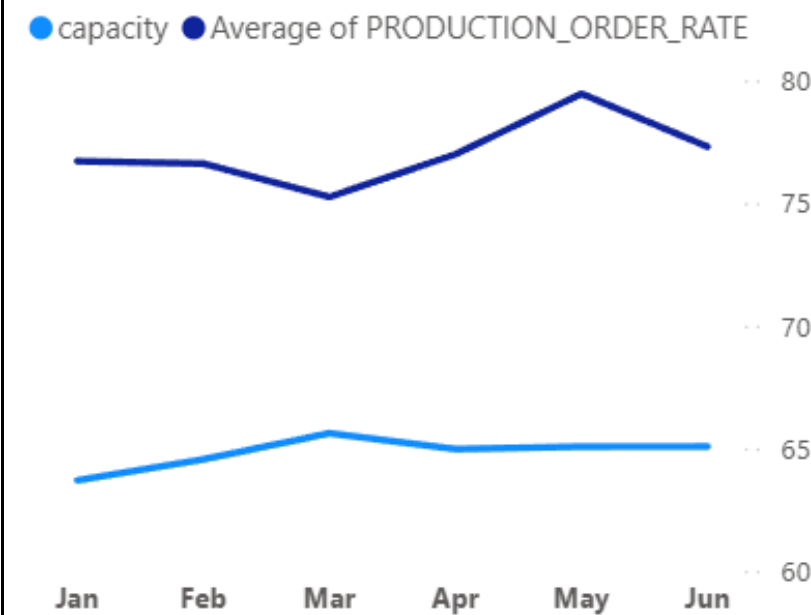
% Change in Expected Output (Month), OEE and KPI_Target_Line by Month



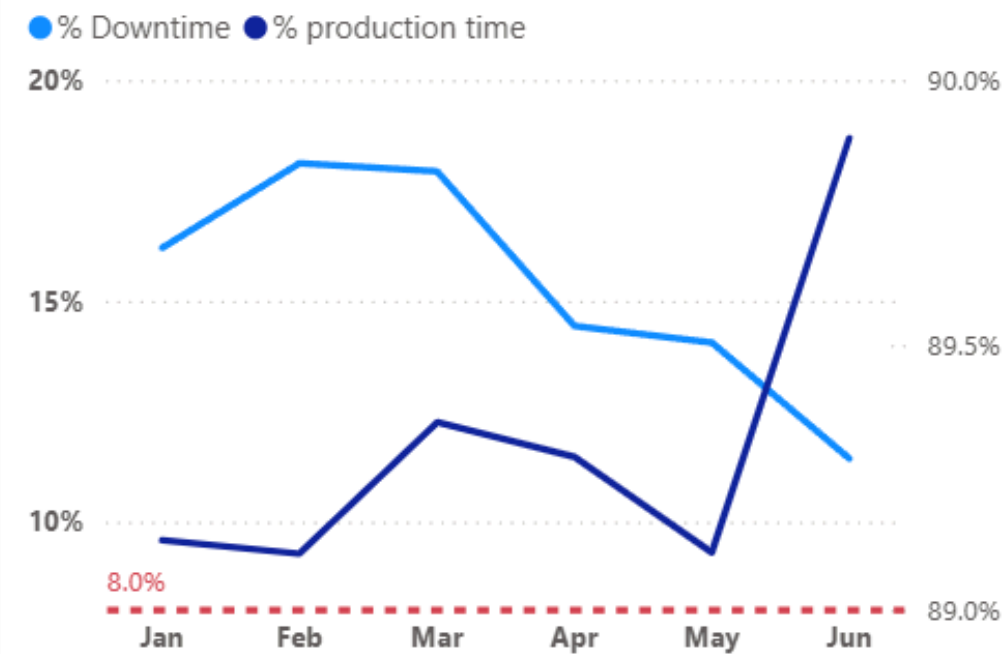
Availability, Performance and Quality by Month



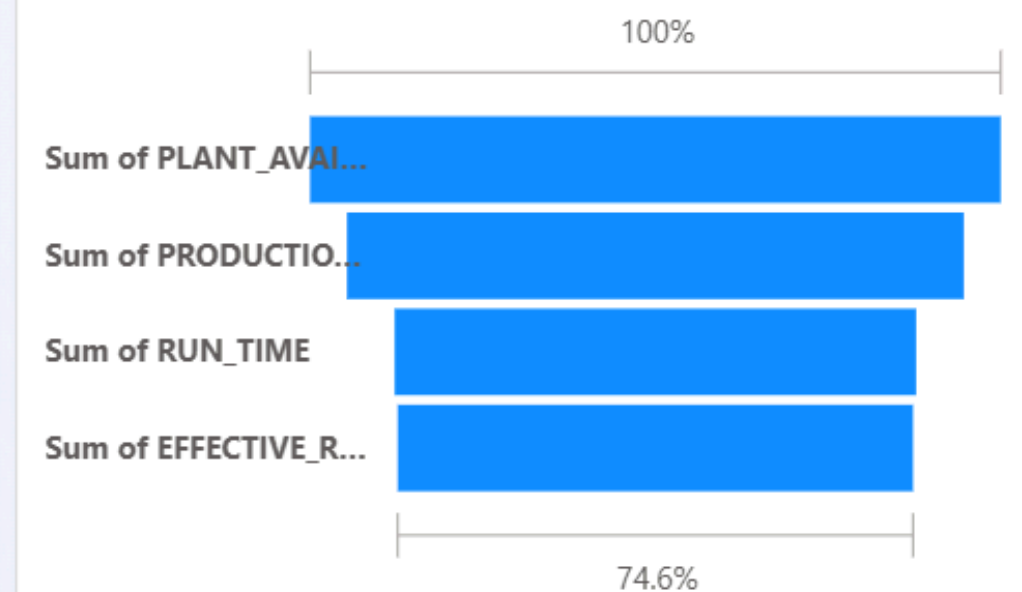
capacity and Average of PRODUCTION_ORDER_RATE by Month



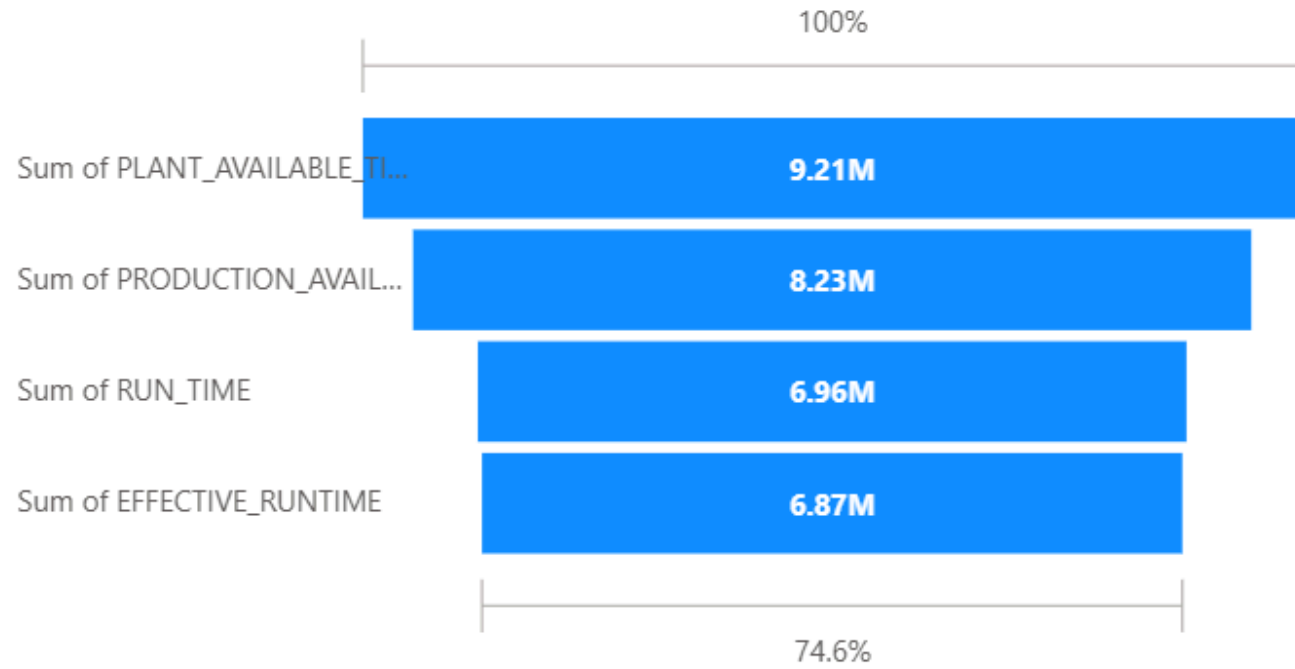
% Downtime and % production time by Month



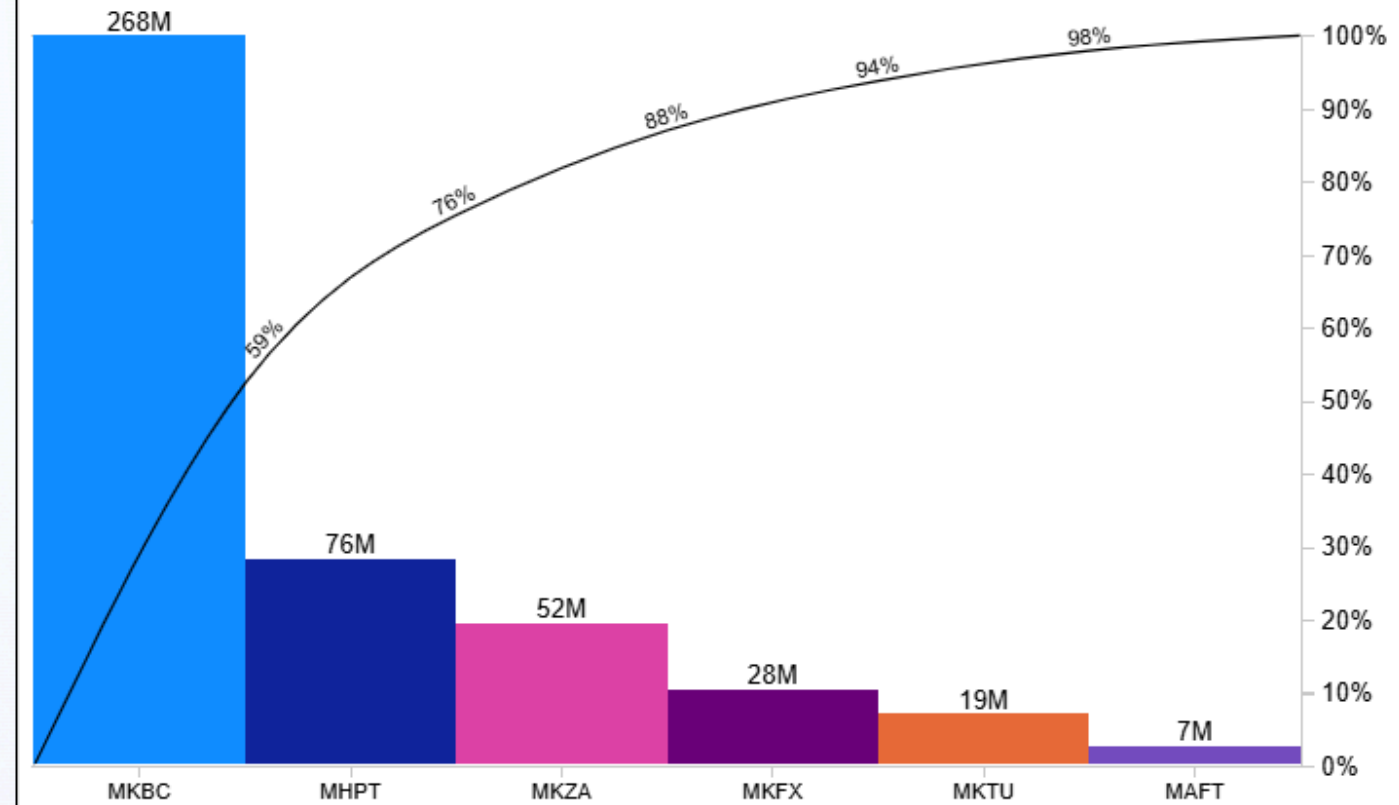
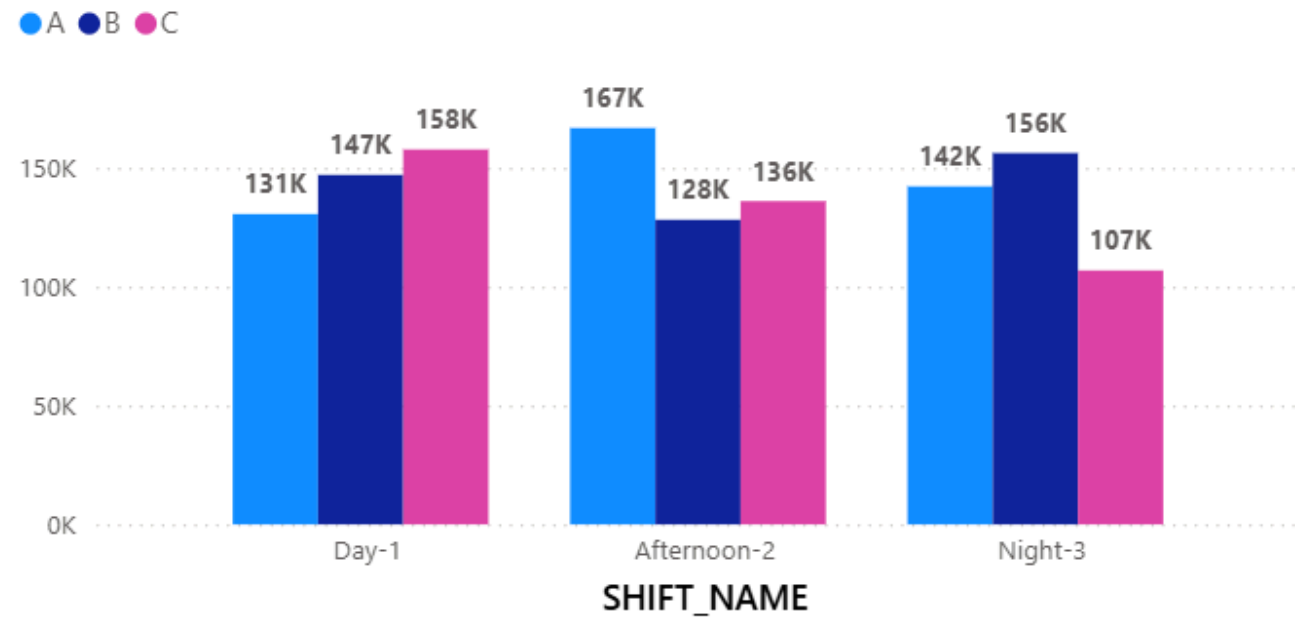
Sum of PLANT_AVAILABLE_TIME, Sum of PRODUCTION_AVAILABLE_TIME, Sum of RUN_TIME and Sum of EFFECTIVE_RUNTIME



Sum of PLANT_AVAILABLE_TIME, Sum of PRODUCTION_AVAILABLE_TIME, Sum of RUN_TIME and Sum of EFFECTIVE_RUNTIME

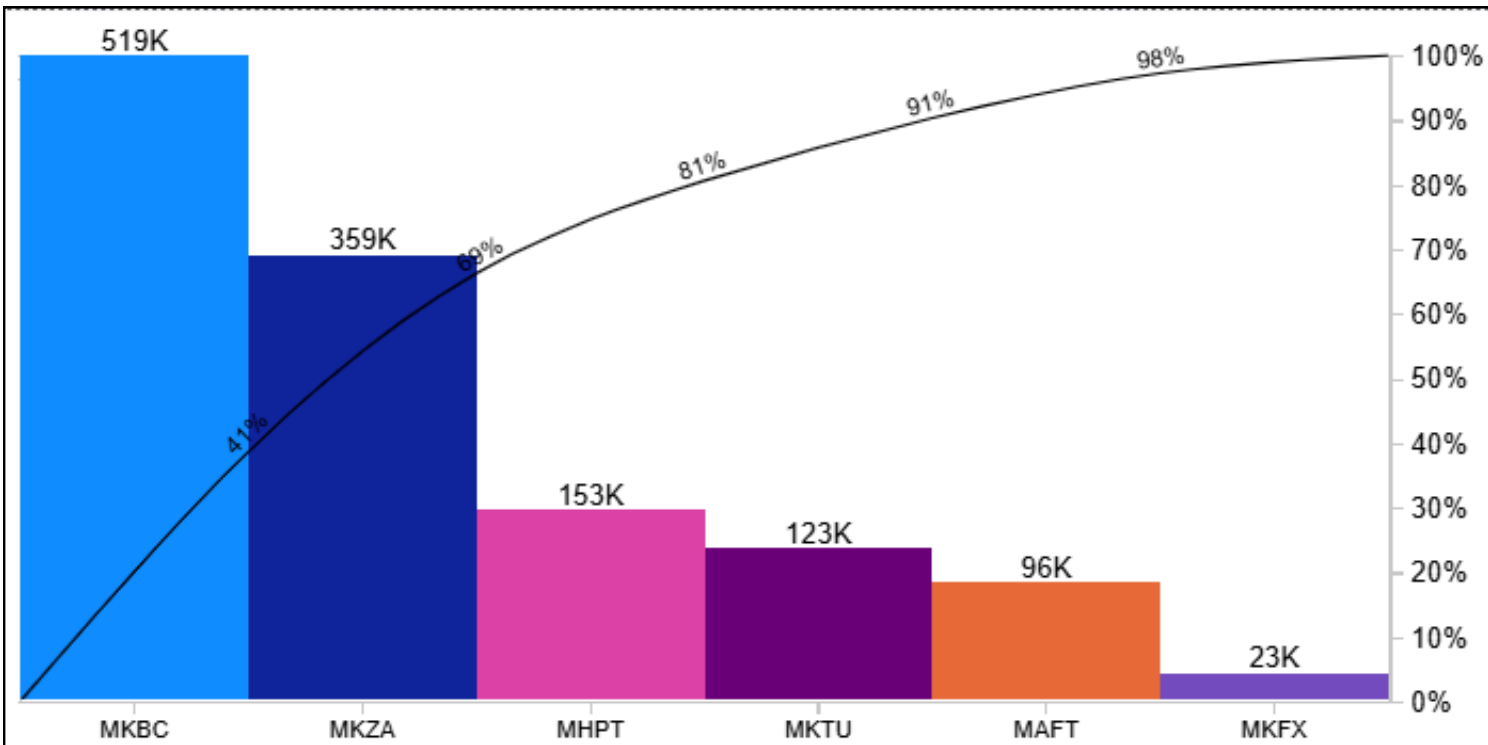


Sum of DOWNTIME by SHIFT_NAME and CREW_ID

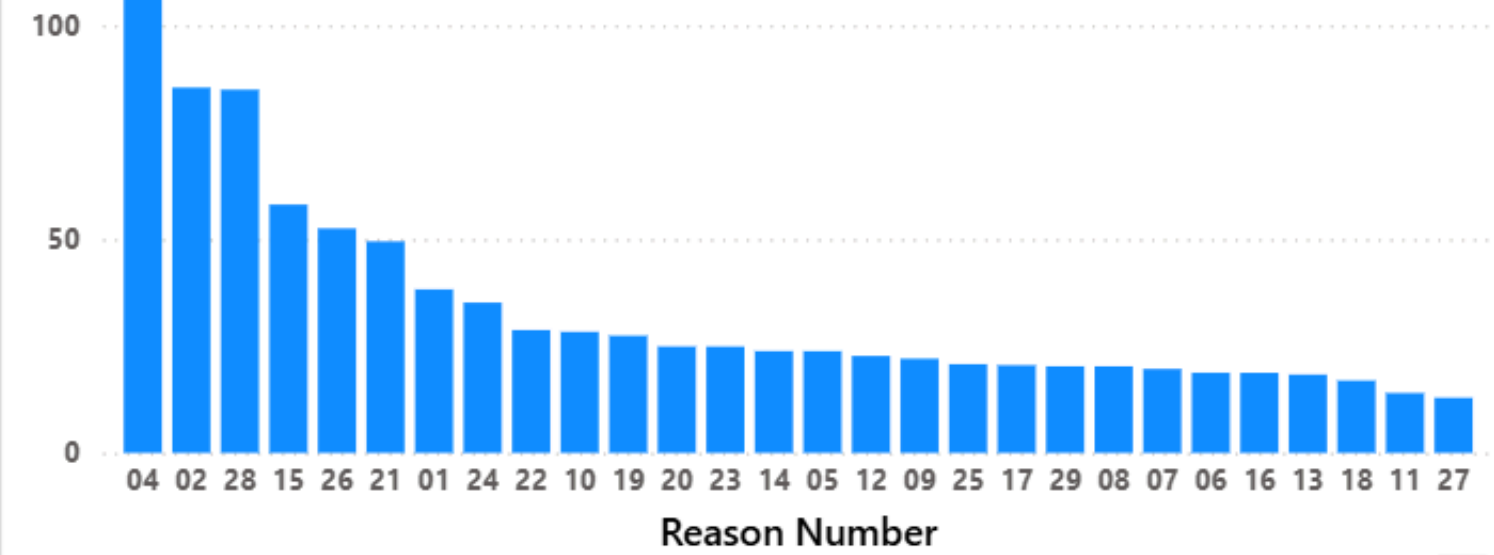


Line code	%GT Sum of total product	%GT Sum of DOWNTIME
MKBC	59.43%	40.77%
MKZA	11.61%	28.20%
MHPT	16.87%	12.06%
MKTU	4.25%	9.66%
MAFT	1.63%	7.54%
MKFX	6.22%	1.78%
Total	100.00%	100.00%

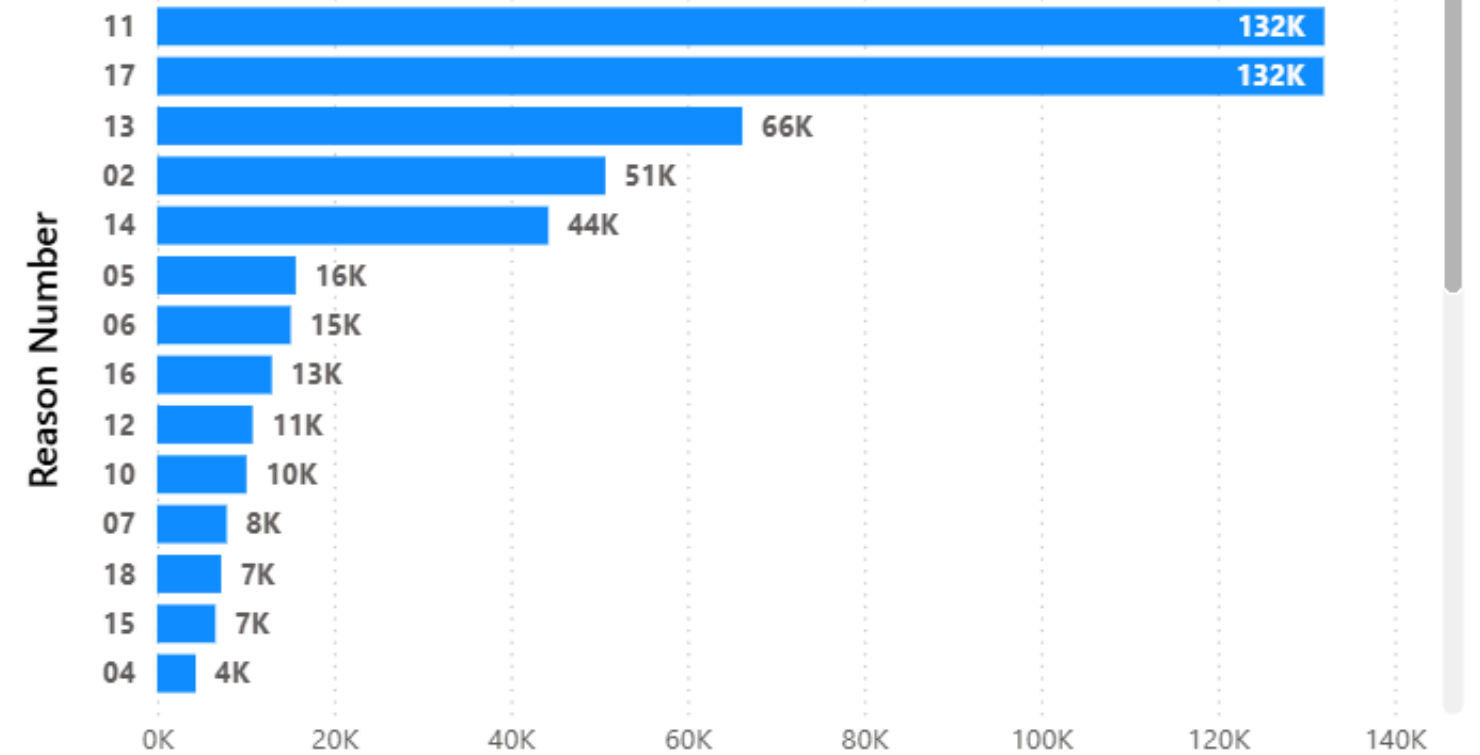
APPENDIX



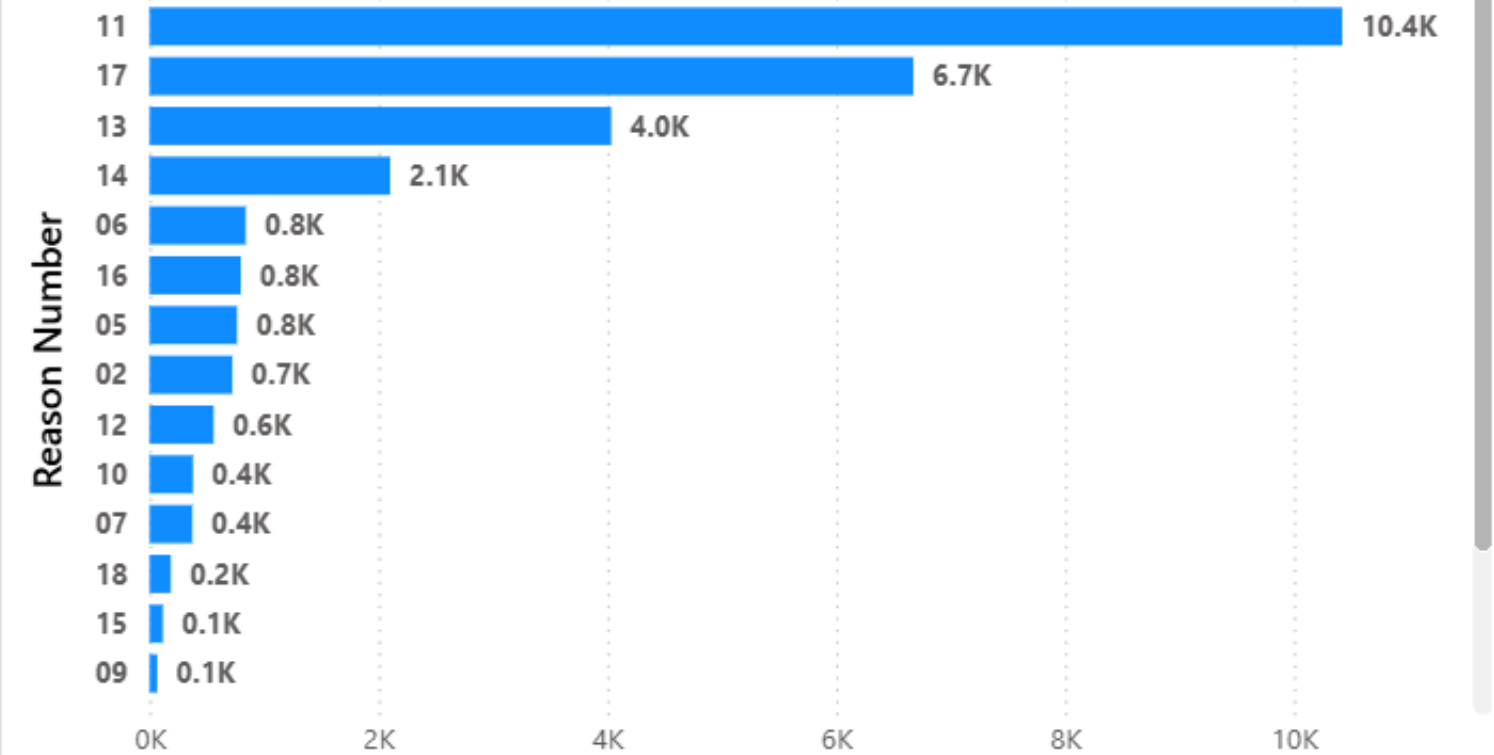
avg downtime each stop by Reason Number



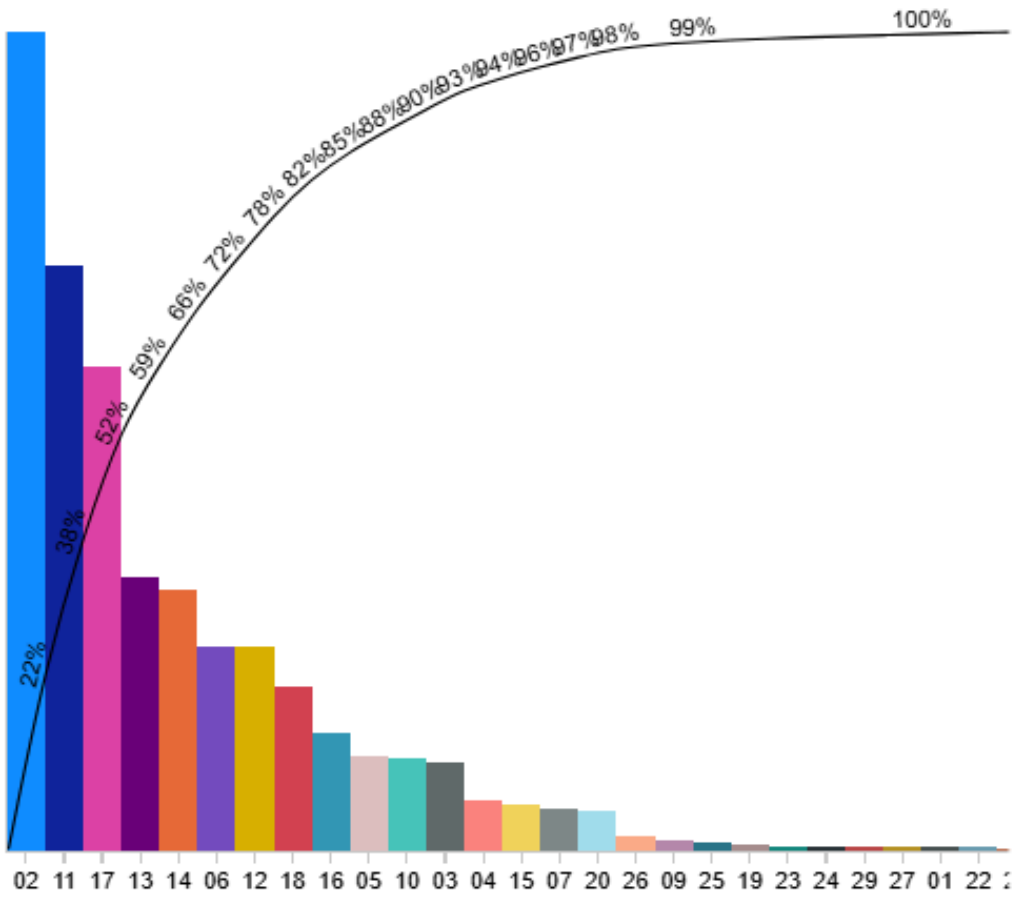
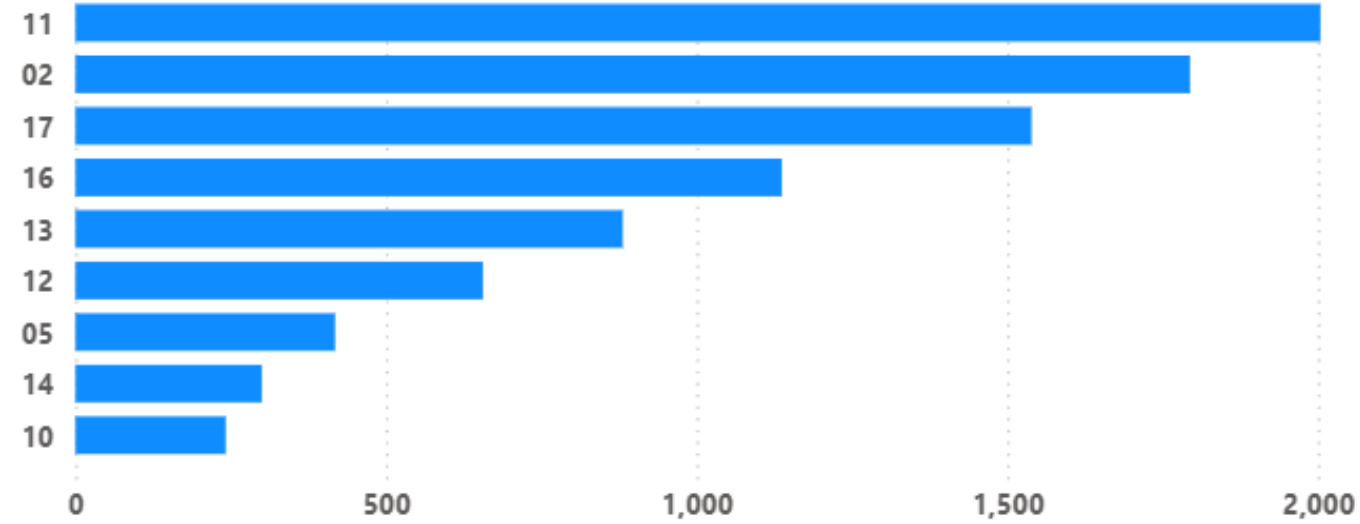
Sum of DOWNTIME by Reason Number



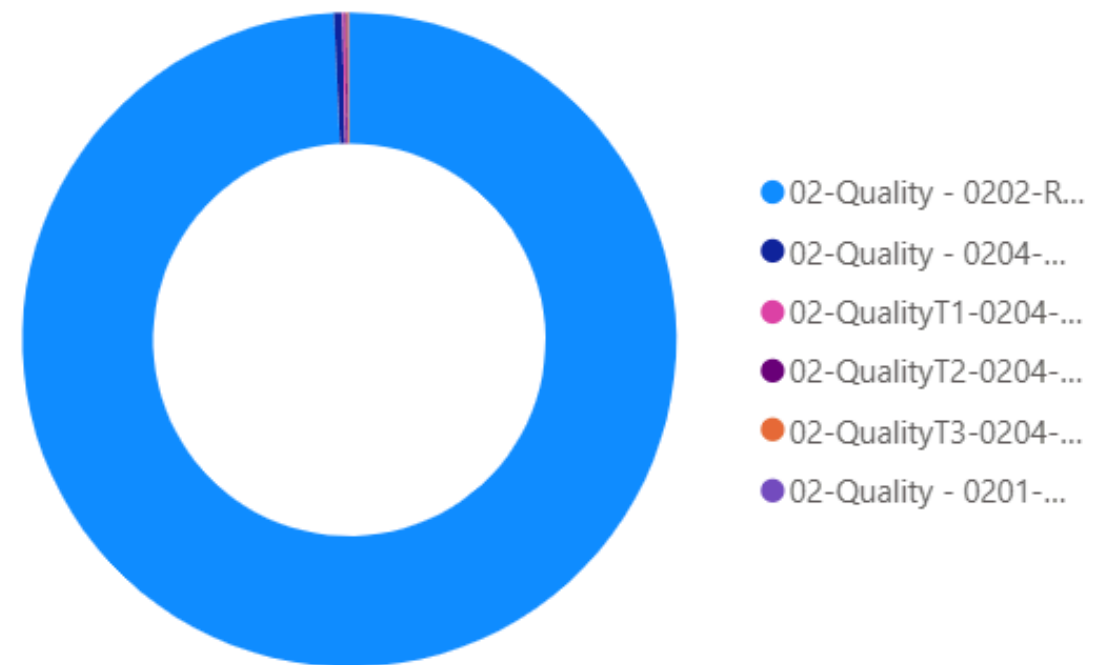
Count of Reason Number by Reason Number



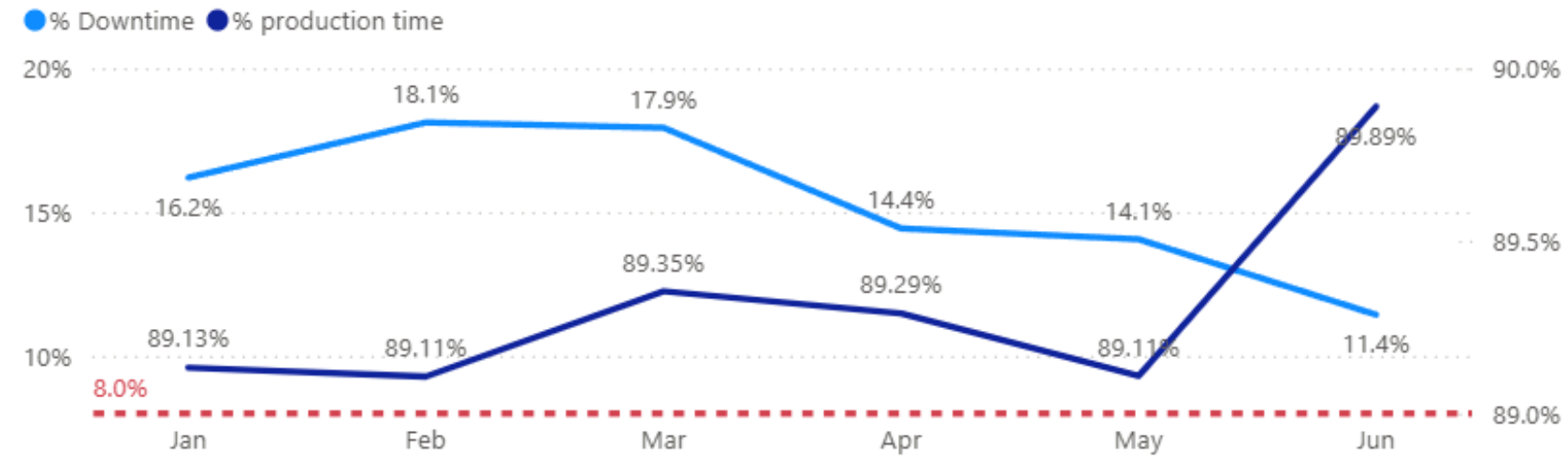
Count of Reason Number by Reason Number



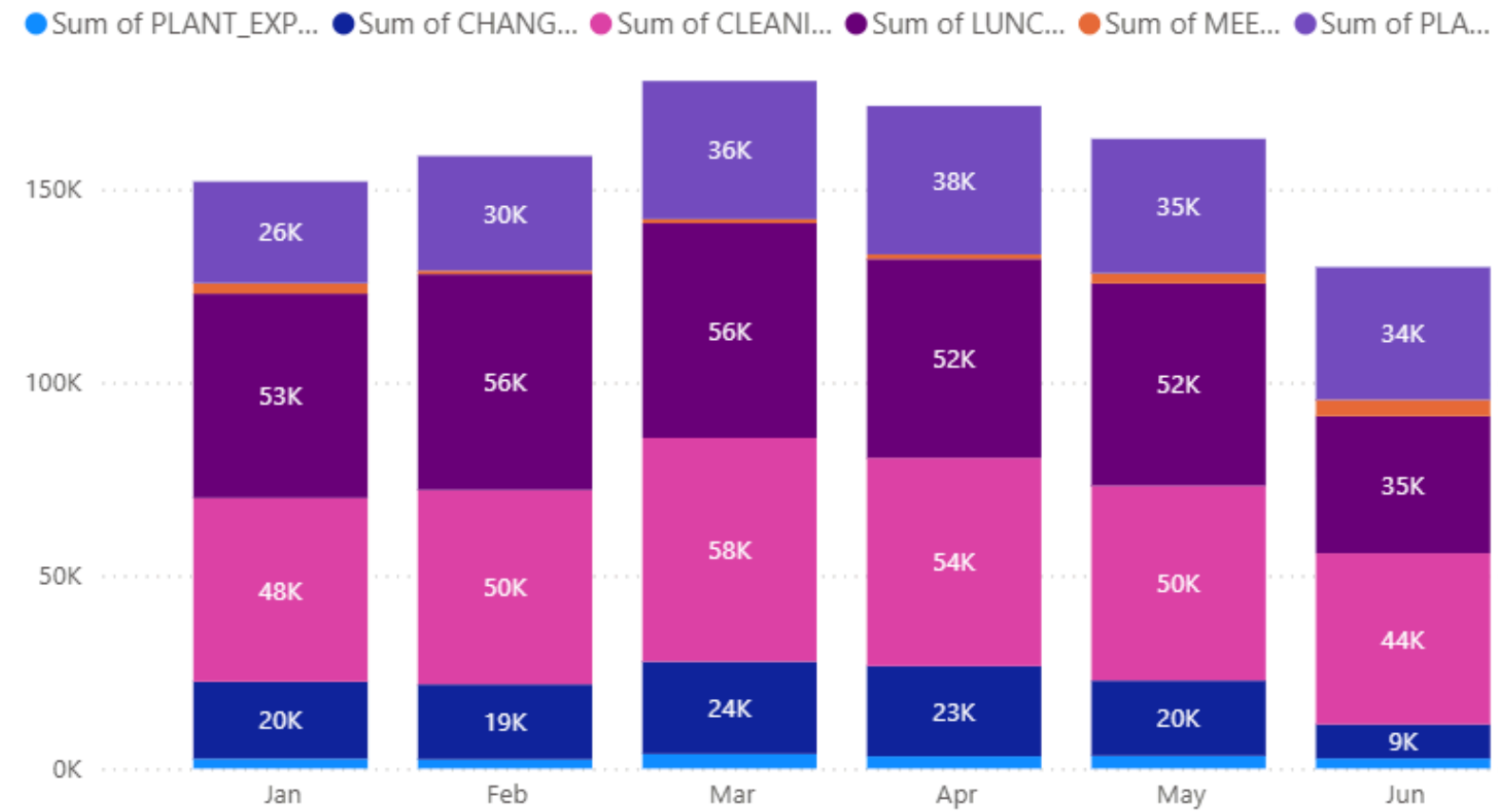
Sum of DOWNTIME by UTIL_REASON_DESCRIPTION



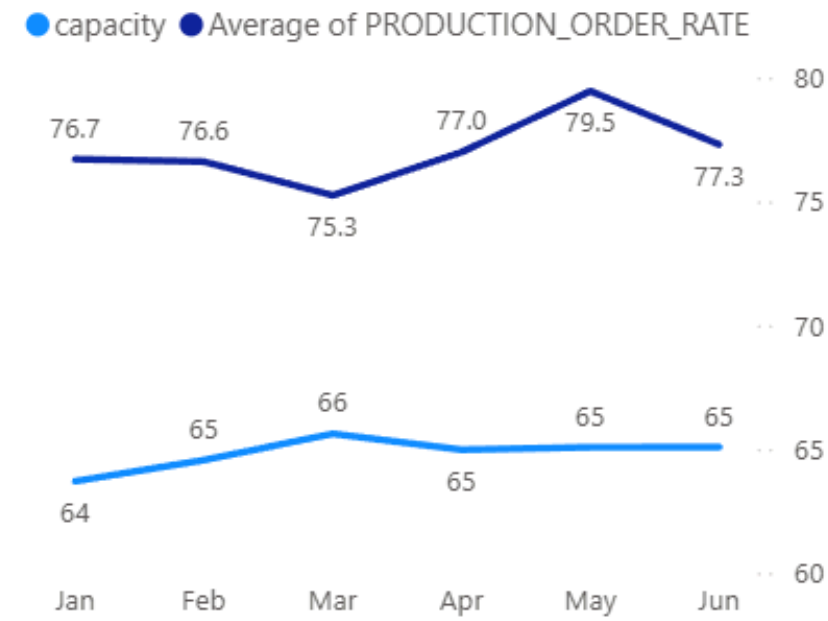
% Downtime and % production time by Month



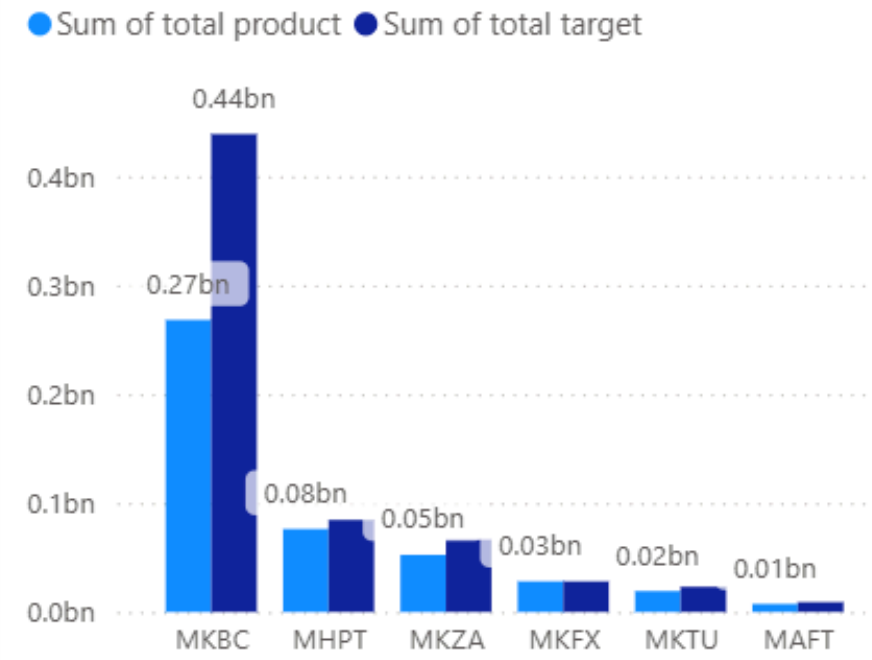
Planned Stops Downtime



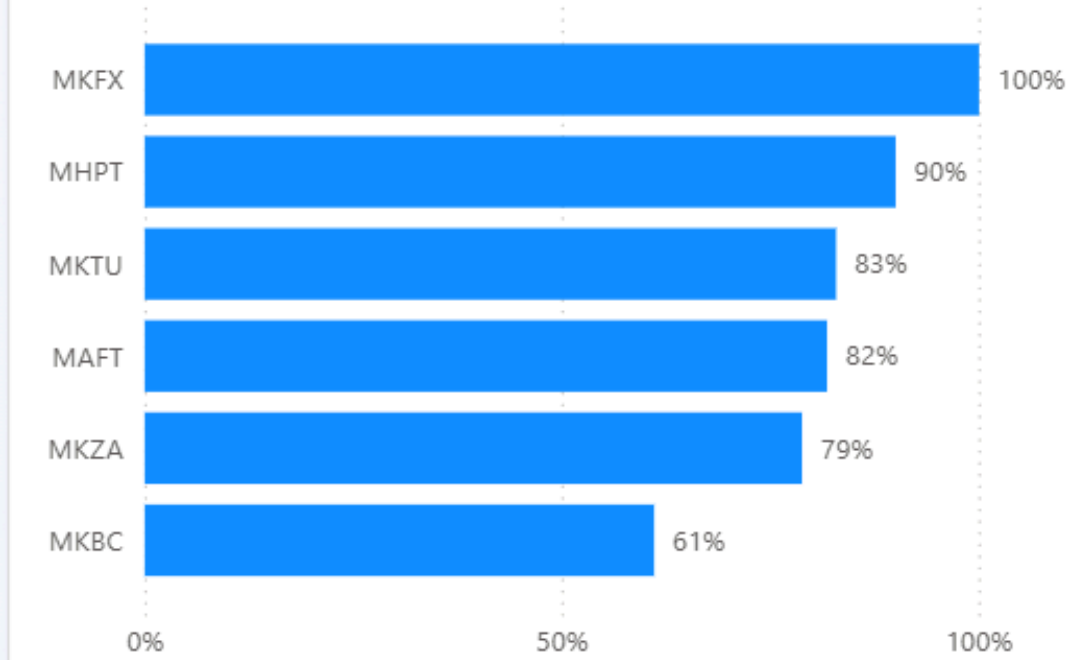
capacity and Average of PRODUCTION_ORDER_RATE by Month



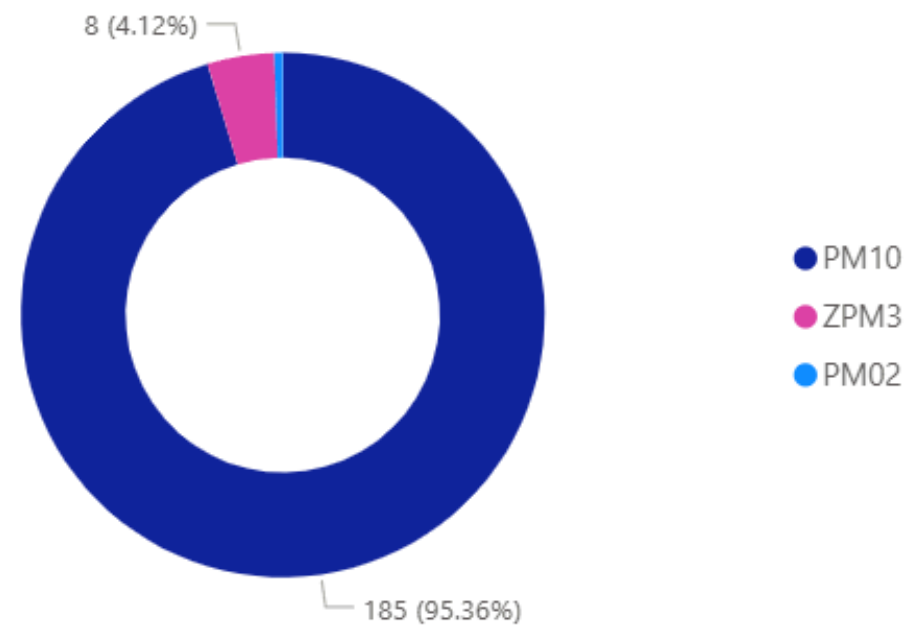
Sum of total product and Sum of total target by Line code



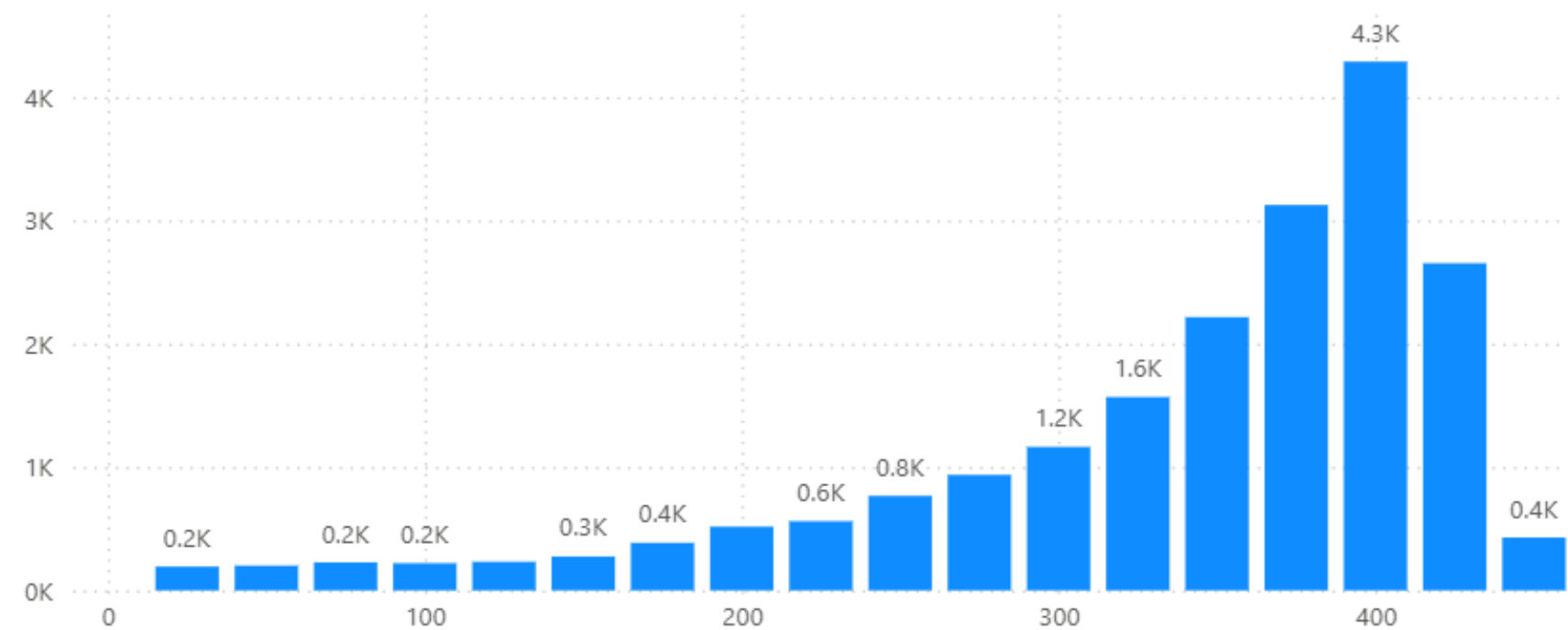
% expected by Line code

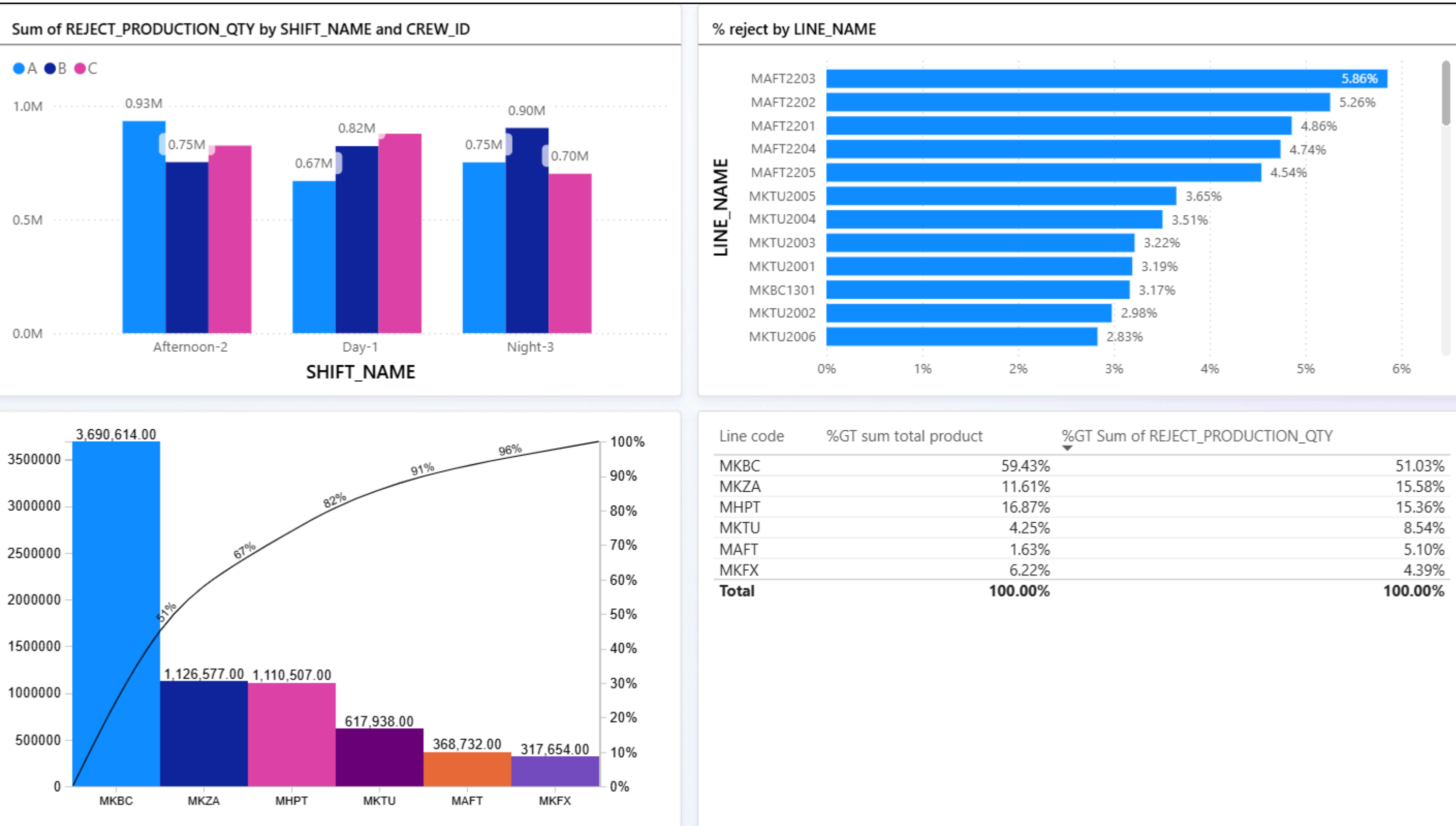


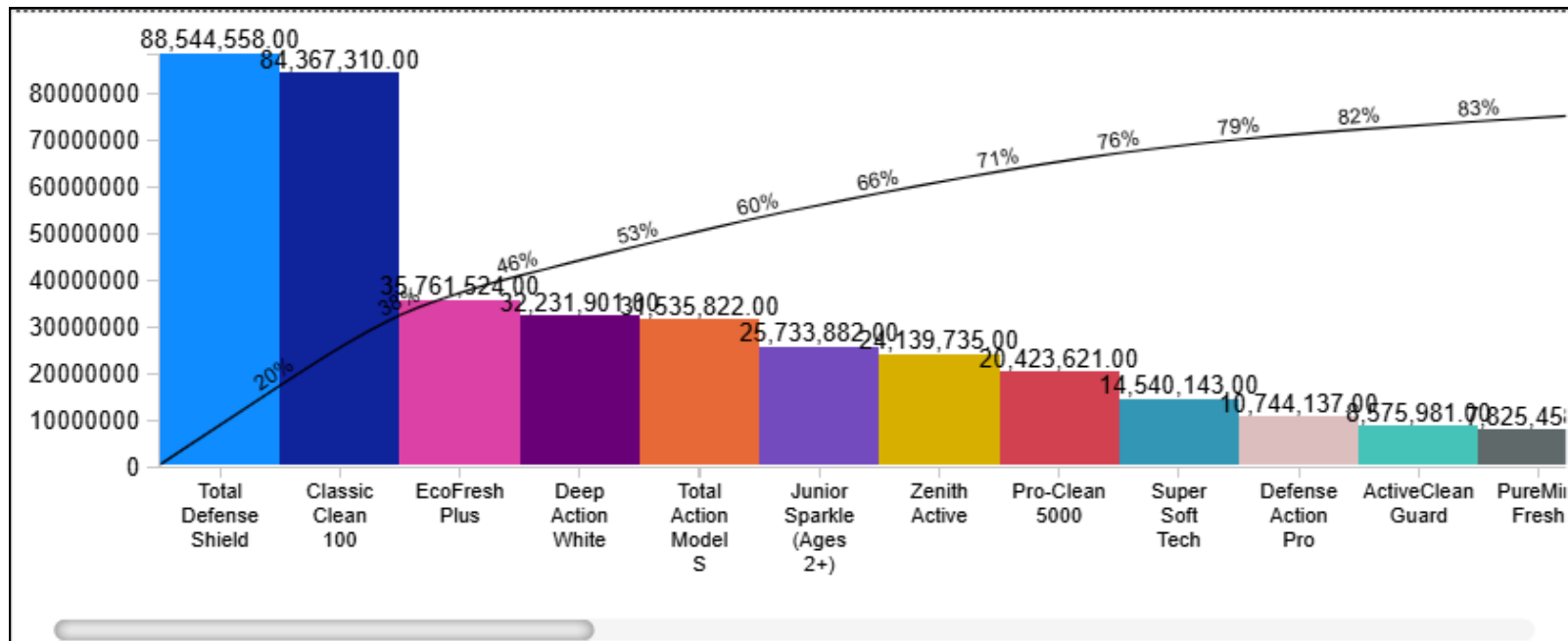
Count of ORDER by ORDER_TYPE



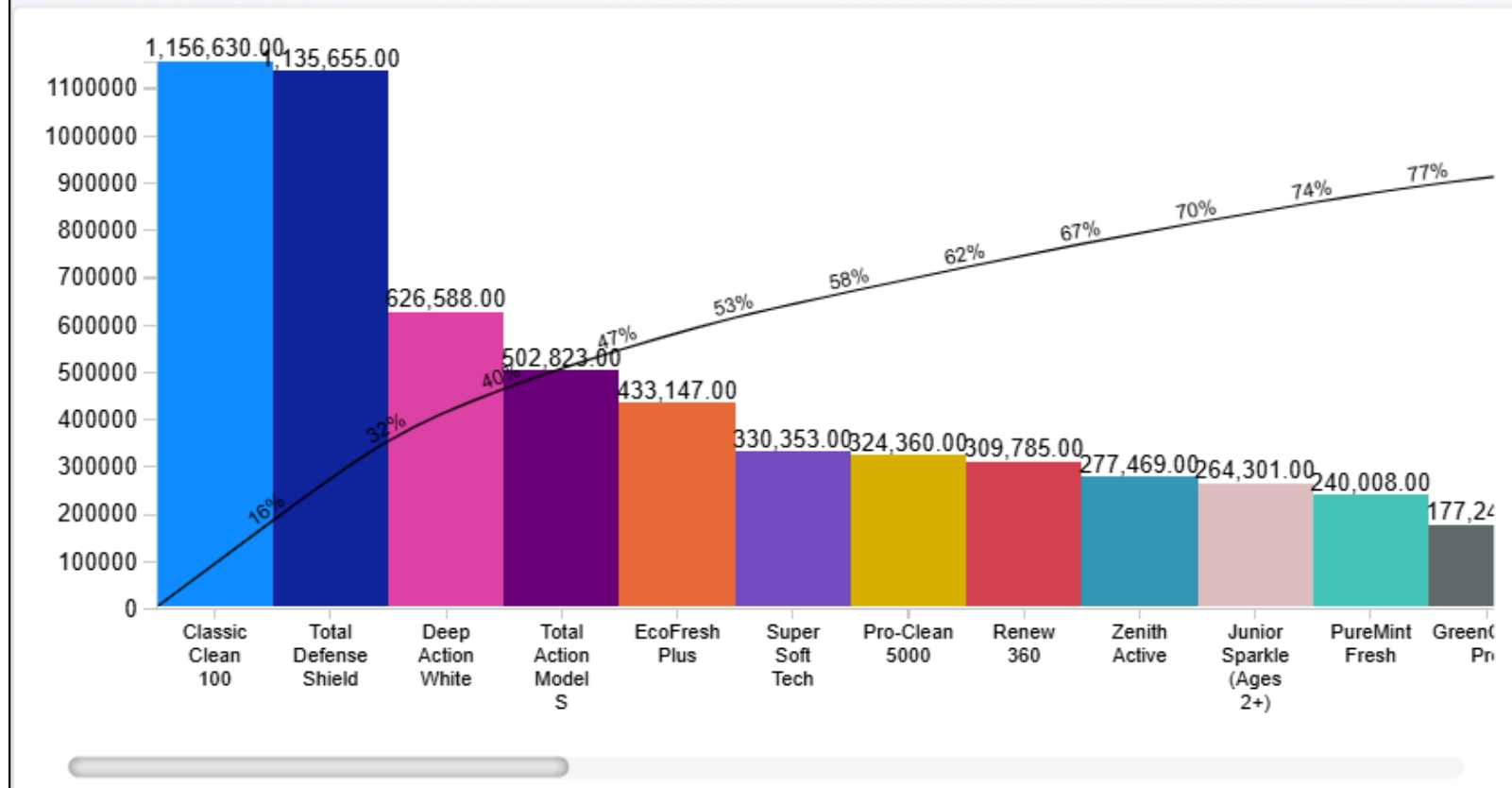
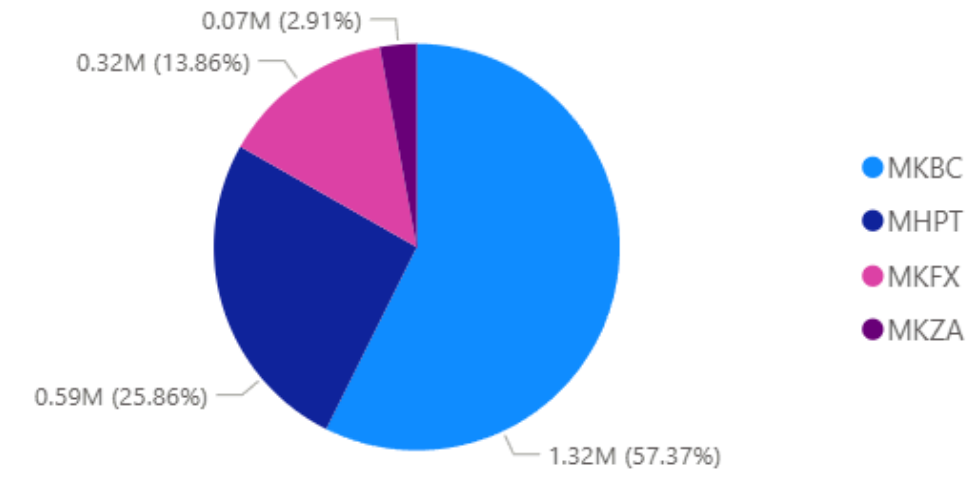
Count of RUN_TIME by RUN_TIME (bins)



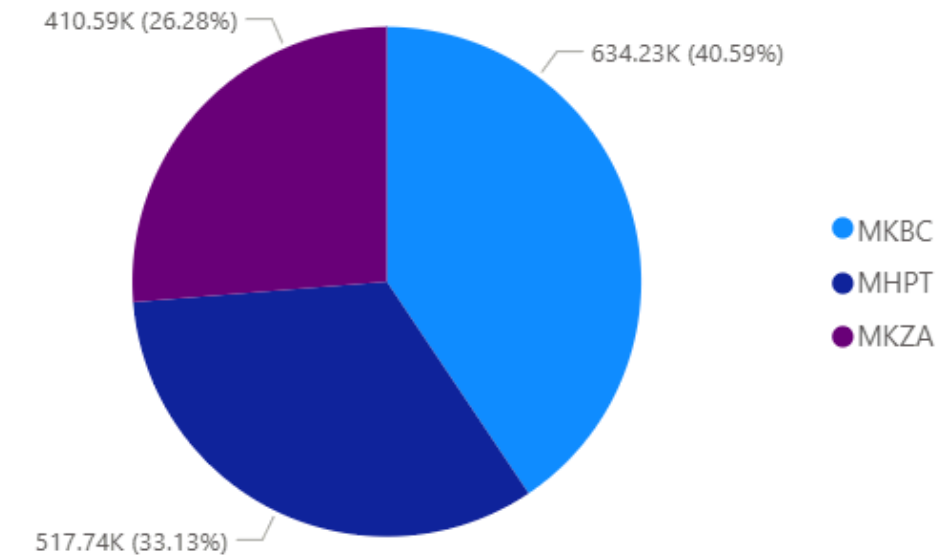




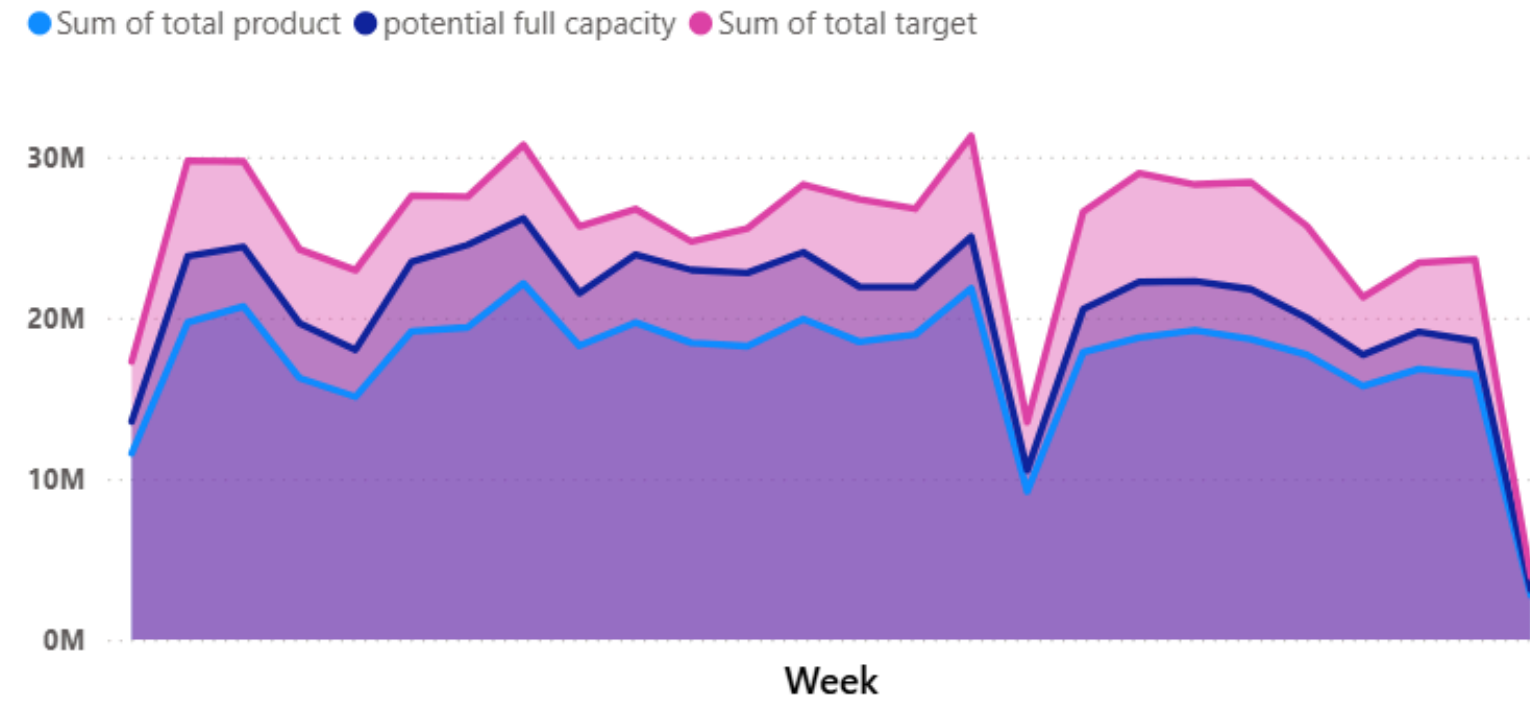
Sum of REJECT_PRODUCTION_QTY by Line code



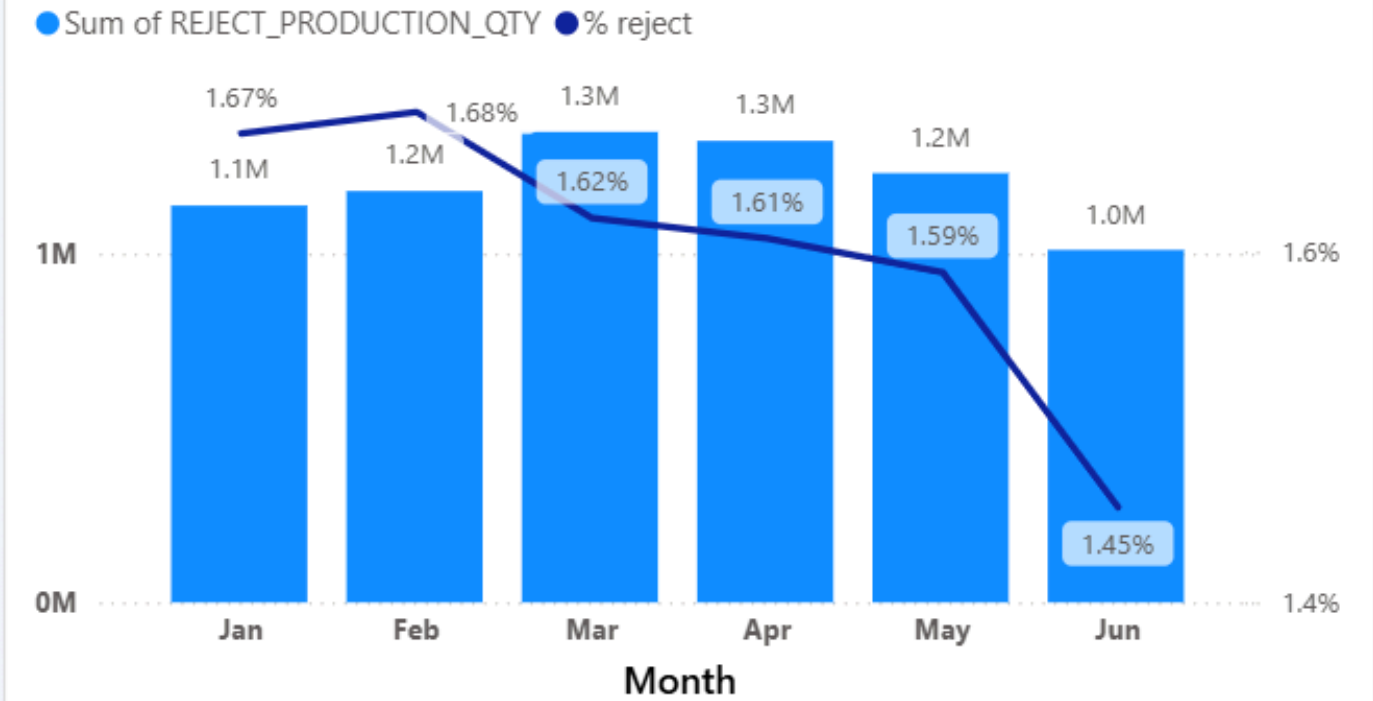
Sum of REJECT_PRODUCTION_QTY by Line code



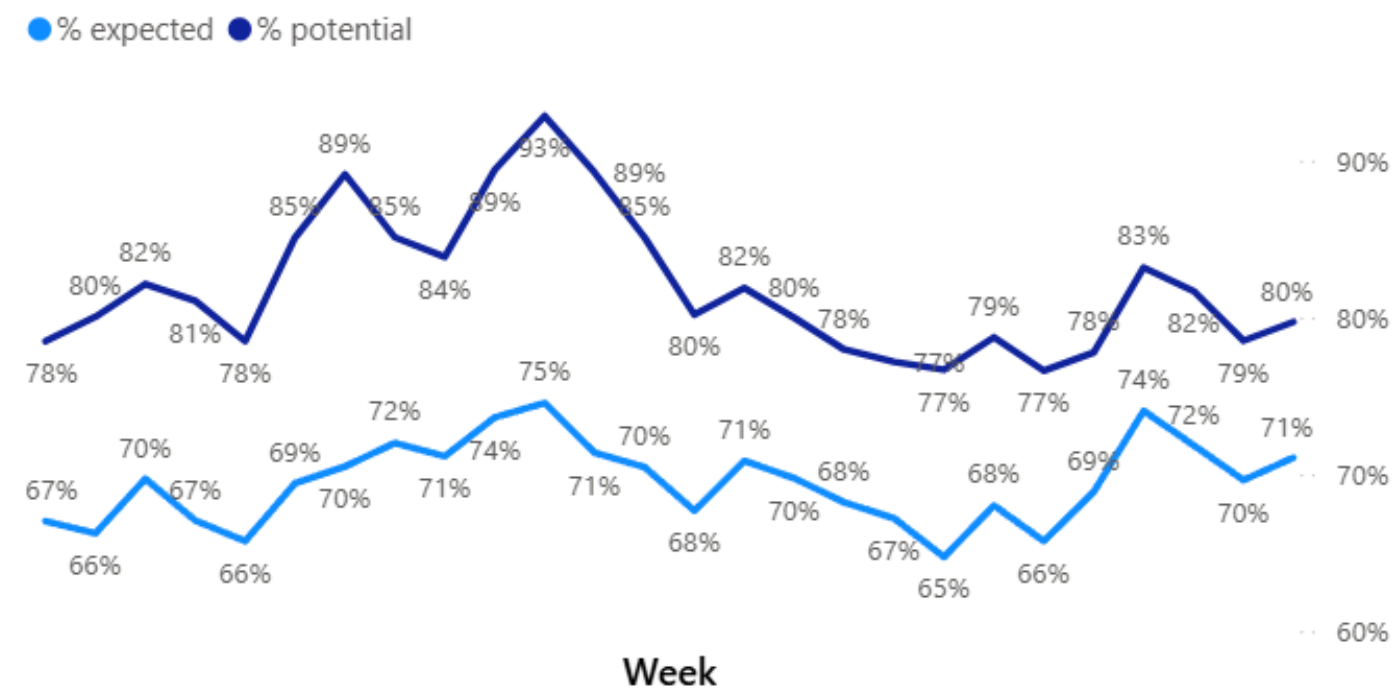
Sum of total product, potential full capacity and Sum of total target by Week



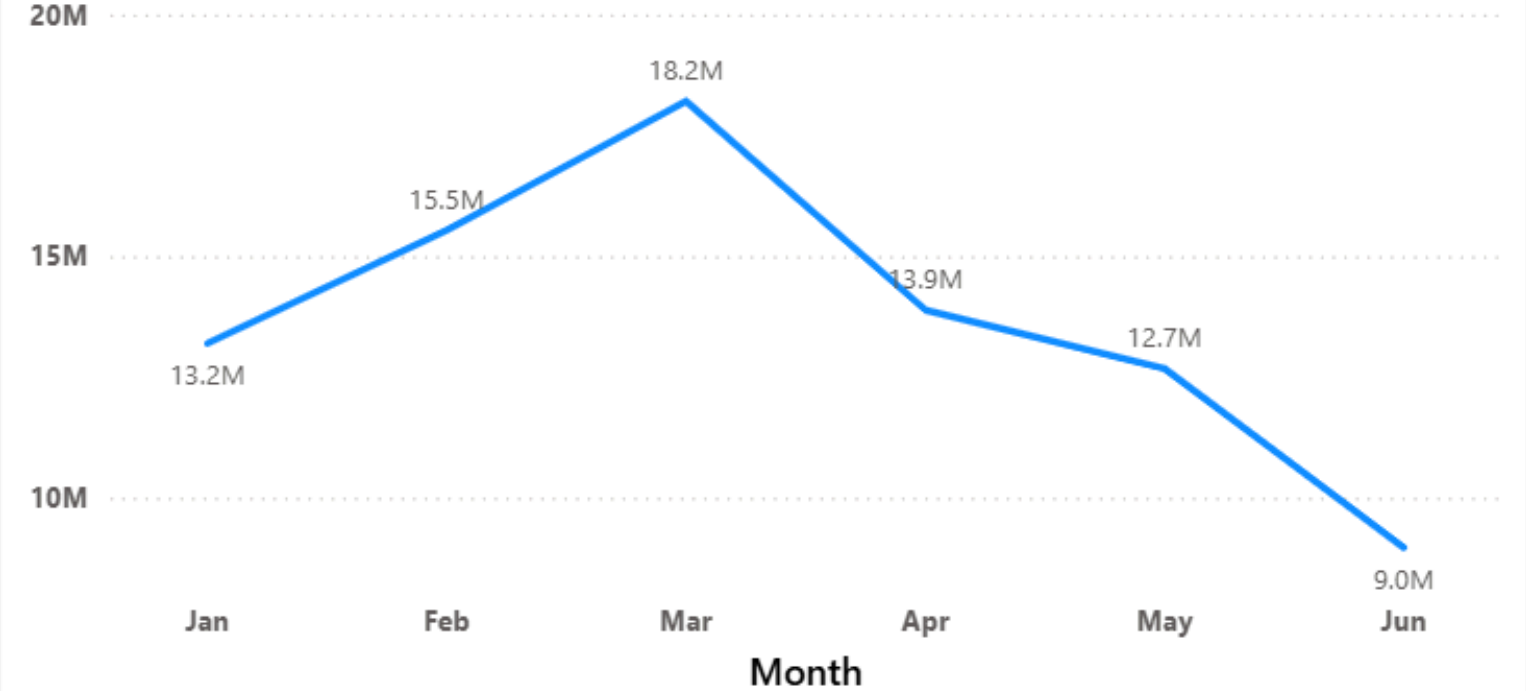
Sum of REJECT_PRODUCTION_QTY and % reject by Month



% expected and % potential by Week

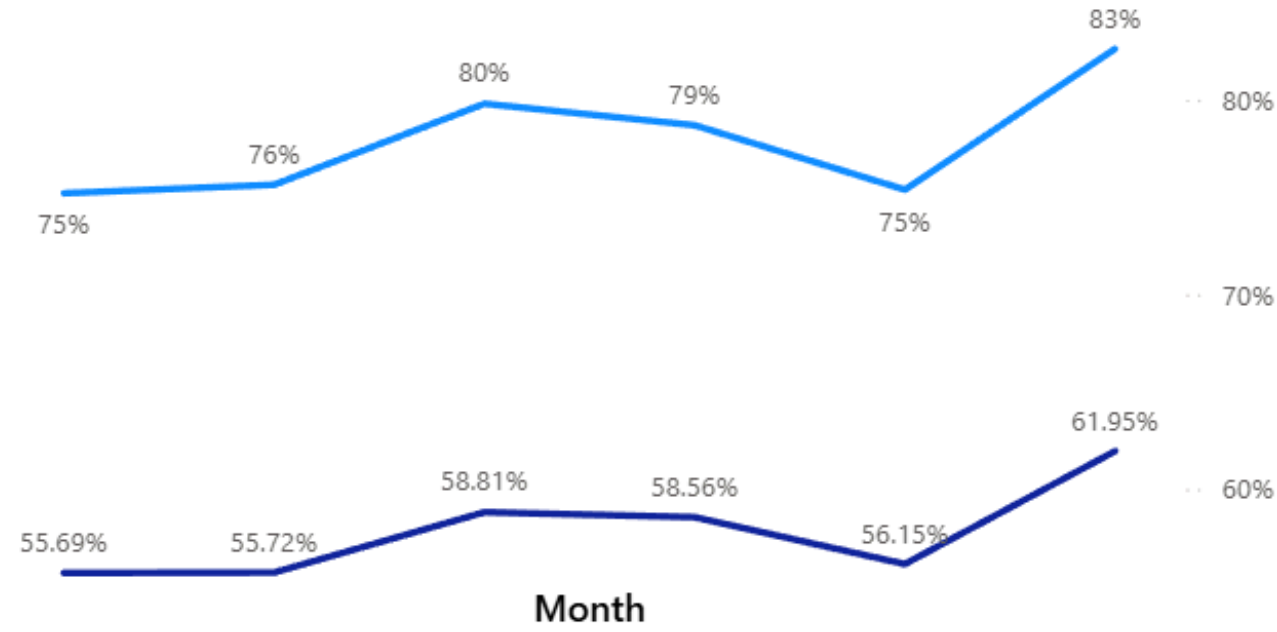


potential capacity by Month



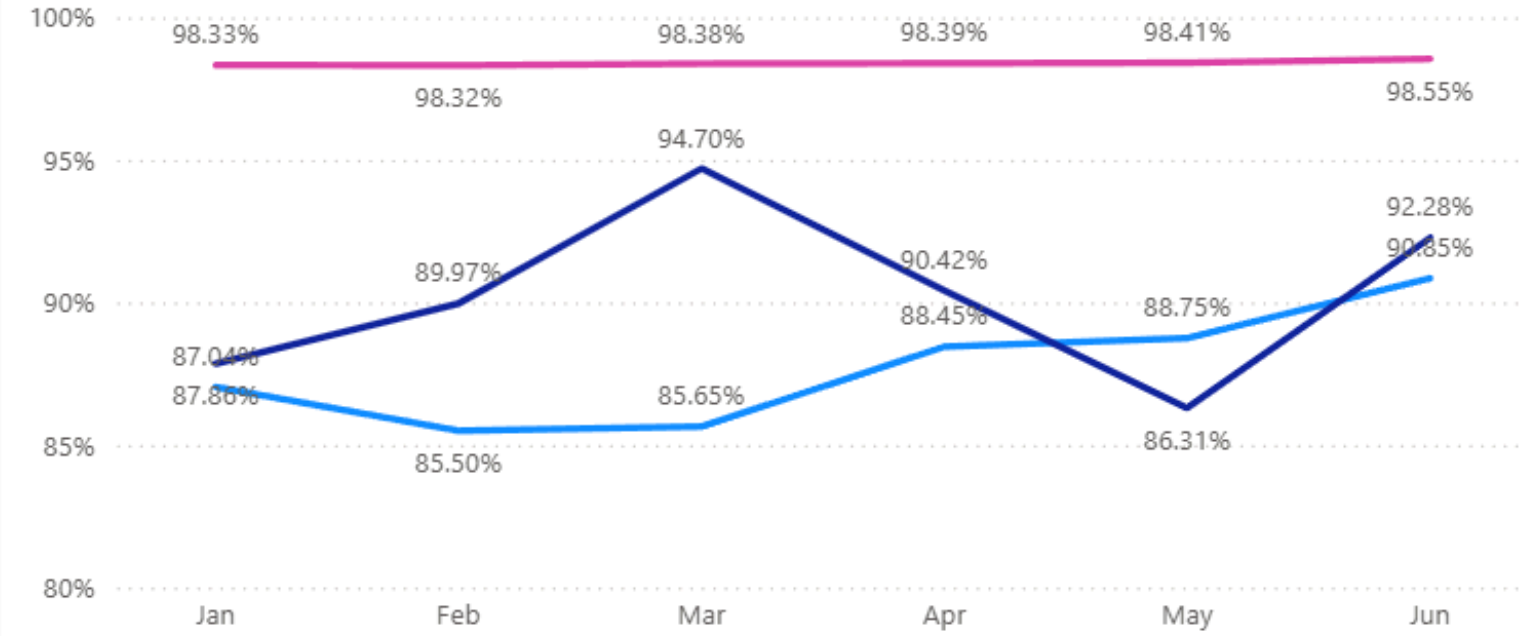
expected OEE and origin OEE by Month

● expected OEE ● origin OEE



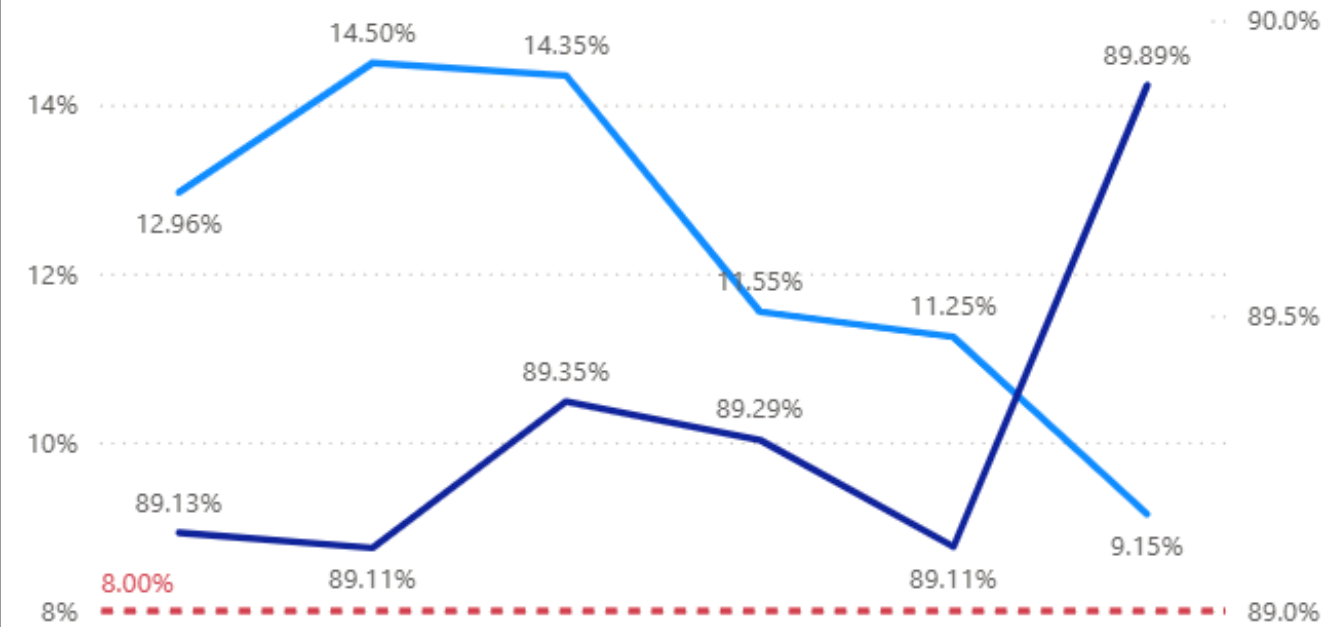
expected availability, expected performance and Quality by Month

● expected availability ● expected performance ● Quality



expected % Downtime and % production time by Month

● expected % Downtime ● % production time



Availability, Performance and Quality by Month

● Availability ● Performance ● Quality

